
Interactive Tutoring for Learning Lithuanian as L2: CEFR-based Prompting Approach

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Abstract

In the educational context, Large Language Models (LLMs) provide a promising approach for the development of tools that are targeted to an enhancement of personalized learning experiences. Given the potential benefits, more research is being placed on the development of LLM-based tutoring systems. The tutoring part is particularly important for successful second language (L2) learning, as it enables the learner with opportunities to receive the targeted learning input and feedback, and to develop the language abilities through interaction with the tutor. Within the LLM-based tutoring system, the language learners would interact with LLMs directly; as such, it is necessary to investigate the LLM behavior over multiple conversational turns. Additionally, it is crucial for language models to be able to adapt to the proficiency level of the learner.

The goal of the thesis is to evaluate whether LLMs can potentially take the role of tutors for teaching Lithuanian as L2 through CEFR-based prompting. The focus is placed on investigation if, or to what extent, the LLMs can be constrained to a specific CEFR level and whether the constraints persist throughout the interaction with the student. The study evaluates whether the instruction-tuned Gemma3 4B, 12B, and 27B models can consistently generate text of increasing text complexity at A1, B1, and C1 language proficiency levels in simulated tutor-student dialogues. The current study is a replication and extension of an already existing experimental design to examine how different model sizes and prompt configurations influence the model's ability to generate text at specific CEFR levels.

Overall, Gemma3 models showed the ability to differentiate between the CEFR levels based on structural surface-level text complexity metrics; however, their ability to consistently maintain the structural complexity of C1 and, in some cases, B1 level text diminished as the conversations progressed, implying that LLMs as tutors cannot reliably generate text suitable for higher-level language learners. Further investigation revealed that such drift in CEFR level alignment is not due to tutor-student LLMs adapting to one another's text complexity; therefore, additional experimental design involving a self-refinement approach to improve the drift in alignment was proposed. While the self-refinement framework improved alignment, the generated output raised unforeseen concerns.

The thesis includes a discussion of the findings, limitations, and proposes further directions for future research. Given the quality of simulated Lithuanian texts and the recurring drift in CEFR-based alignment, it is clear that Gemma3 language models, in their base form, cannot be used as a tool for interactive tutoring of Lithuanian as L2.

Keywords: alignment drift, CEFR, Gemma3, Lithuanian language, language models, second language learning

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1. Introduction

The implementation of Large Language Models (LLMs) in the educational context presents a promising approach to enhance personalized learning experiences. There has been a rapid growth in the use of LLMs in education, progressing from their integration into simple, intelligent tutoring systems to more advanced applications where LLMs end up taking the roles of interactive conversational agents for learning. The biggest advantage the LLMs as conversational agents pose is that they provide the learners with the ability to access the learning environment at any time, allowing the learning in their own style, at their own pace. Moreover, the interaction part of these systems plays a crucial role in the learning domains such as second language (L2) learning (Loewen & Sato, 2018).

Language models are not designed to act as L2 tutoring agents. However, different prompt engineering strategies are increasingly being used to elicit more specialized model behaviours (Giray, 2023; White et al., 2023). In L2 learning adapting, the model's behaviour is especially crucial, as the second language learners vary in their language proficiency. As such, the agent must be able to adapt to the skills that the learners possess. Research on LLM-based L2 learning usually relies on the Common European Framework of Reference (CEFR)-inspired prompting strategies, which define the learner's abilities at different language proficiency levels.

While the CEFR-inspired prompting strategies differ across the studies, investigations are mostly focusing on highly explored languages such as English, Italian, Chinese, or Spanish (Lin-Zucker et al., 2025; Lukešová & Jennings, 2025; Marzona et al., 2025). Almasi & Kristensen-McLachlan (2025) have explored not only whether LLMs as tutors are able to adapt to the specific language proficiency levels, but also how well the adaptation lasts throughout the conversation with a student. However, the latter study focused on LLMs' ability to adapt to the student who is learning Spanish as L2, and therefore, it is still questionable how well the LLMs perform in more under-explored languages, such as Lithuanian.

The present thesis is a partial replication and an extension of the study performed by Almasi & Kristensen-McLachlan (2025), aiming to investigate whether language models, when taking the role of a tutor for Lithuanian as a second language under CEFR-based system prompting, are able to adapt to a specific language proficiency level, and whether the adaptation persists throughout the whole conversation with a student. Moreover, the thesis addresses the limitations identified in the prior work, extending the investigation a bit further.

In section 2, I provide an overview of related work on large language models in the education context. I start by describing what Intelligent Tutoring Systems (ITS) are and what benefits LLMs provide when integrated into these setups. Subsequently, I describe what human-like tutoring is and present previous work on LLMs as second language learning tutors. The latter leads to the

introduction of CEFR and elaboration on using CEFR as a prompting strategy to elicit tutor-like behavior in LLM. Then I introduce the central study of this thesis, which is the one conducted by Almasi & Kristensen-McLachlan (2025), and propose the Lithuanian language as a case study.

In section 3, I discuss the process of language model selection for the simulation of tutor-student dialogues. I describe the experimental design, such as the setup of inference, simulation of tutor-student conversations, and the system prompts used. As selected LLMs produced a noisy output, I present the steps of data preprocessing. I discuss the methods for the analysis of the LLM output, such as the computation of relevant metrics and the evaluation of the LLM text against the reference data. Comparing the simulated data against the reference is an essential part of the thesis, as it provides an overview of whether LLM-generated texts differ from pedagogically designed L2 learning materials for learners at specific language proficiency levels. Since no official benchmarking dataset is available, I present the process of how such a dataset was curated for the current study, and describe it based on morphological, lexical, and syntactic feature frequencies across CEFR levels. Then, I describe the linear mixed-effects analysis, which is applied to structural text complexity metrics extracted from the LLM-generated texts. Specifically, text length and mean dependency distance are used as surface-level metrics of structural text complexity to examine whether the generated outputs differ significantly across CEFR proficiency levels. Subsequently, I present an additional analysis of the simulated tutor-student conversations to gain further insight into the interactional dynamics observed in the generated tutor-student conversations. Lastly, a new experimental design is proposed to evaluate whether the LLMs' output can be improved once the model is instructed to self-refine.

In section 4, I present the results of the analysis, both qualitative and quantitative. The qualitative part involves the discussion on the quality of the LLM-generated text before the data was pre-processed. Then I briefly present the errors of lexical validity found in the simulated texts, which describes how limited the selected LLMs are in generating valid words. The quantitative part of the results section presents the findings relevant to the analyses described in Section 3.

The remainder of the paper discusses the results with respect to what has been found and what they entail, including the limitations, ethical considerations, and suggestions for future work.

2. Related work

Research in artificial intelligence has been significantly shaped by the development of Large Language Models (LLMs). Due to the rapid advancements, LLMs have been widely integrated into various application domains. In e-commerce, for instance, LLMs might be used to facilitate the process of creating the descriptions of the products or to summarize customer reviews (Palen-Michel et al., 2024). In healthcare, LLMs are used to facilitate diagnostic decision-making processes, enhance communication with patients (Bélisle-Pipon, 2024), or answer questions related to competency development (Unell et al., 2025). The latter use of LLMs extends beyond the medical context, and researchers are increasingly exploring their role within education more broadly (Alhafni et al., 2024).

2.1 LLMs in Education

Within the broader educational context, researchers have explored the capabilities of LLMs for analytical tasks such as text classification. For instance, LLMs have been utilized to perform a sentiment analysis of student-written comments on social media platforms, by assigning them to a category of psychological distress (Pilicita & Barra, 2025). The results showcased that LLMs achieve a promising accuracy of the emotion classification, therefore, indicating that models could potentially provide significant insights into students' well-being, allowing the teachers to assist them with targeted, adaptive support. However, the investigation of the role the LLMs can take within the education is not limited to the analytical applications alone. In fact, although LLMs can provide valuable analytical and adaptive insights, they are becoming more and more conceptualized as personalized learning agents (Oprea & Bâra, 2025).

The potential of Generative Artificial Intelligence (GenAI) tools, such as AI-powered chatbots like ChatGPT, to enhance the targeted learning experiences has already been acknowledged. With such tools, students can engage in a more dynamic learning: students are enabled to explore various topics, ask questions about them, receive timely, immediate answers or feedback (Melzer et al., 2025). Moreover, LLMs show a strong potential to be used as adaptive learning tools that tailor educational learning content to the individual learner's pace (Albdrani & Al-Shargabi, 2023), needs, and the learning style itself, resulting in enhanced overall learning outcomes (Kaddour et al., 2023; Melzer et al., 2025).

2.2 LLMs within Intelligent Tutoring Systems

2.2.1 Intelligent Tutoring Systems

Given the potential benefits that LLMs can offer within the educational context, more attention and research are placed on **LLM-based Intelligent Tutoring Systems (ITS)** development. The early architecture of such software, together with the integration of AI techniques, has created the capabilities for the system to track the learning the student is engaging in, provide hints throughout

the whole learning process, and deliver feedback. Since the 1970s, it has been argued that ITS has to consist of the three distinct components: the first one should track the knowledge that the student possesses, the second one should contain the knowledge of the domain that the student is learning, and lastly, the third one should consist of the knowledge with the strategies for teaching (Nwana, 1990).

Such a standard ITS with rule-driven algorithms relies on predefined learning materials and domain knowledge. However, the domain knowledge encoded in these systems is not enough - it comprises the knowledge of an expert within a specific domain, which might be a challenge to translate to a student without utilizing appropriate, adaptive explanations. Additionally, extending the system with more domain knowledge requires manual adaptations, which makes the scalability, material adaptability, and the overall maintenance of these systems inefficient and time-consuming (Z. Liu et al., 2025).

2.2.2 From Traditional to LLM-based Intelligent Tutoring System

To overcome the drawbacks of the standard ITS, Liu et al. (2025) have proposed an idea of an LLM-based intelligent tutoring system. In this case, rather than relying on predefined knowledge or domain representations, this part of the system could be replaced by the LLM. The greatest benefit of the LLM in this case is an extensive knowledge across various domains acquired during the model's pre-training. As such, the manually encoded, static knowledge pre-defined in the ITS could be replaced by a dynamic alternative. Furthermore, the authors have introduced retrieval-augmented generation (RAG) and prompt engineering to improve the system as a whole.

RAG is a framework that is designed to improve the output that LLMs can generate by making it possible for LLMs to retrieve the data based on the sources that the model has possibly not been trained on (Arslan et al., 2024). As a result, during the inference, LLM is capable of dynamically retrieving domain-relevant documents and generating text output based only on relevant information. Such an addition to the domain knowledge component within the standard ITS significantly improves the maintainability and accessibility of knowledge. By utilizing the external domain, knowledge LLMs have been found to generate less hallucinations and generally enhance the quality of text generation (Lewis et al., 2021).

In addition to RAG, Liu et al.'s (2025) study has utilized prompt engineering as another tool to control the way the LLM generates text. Instead of fine-tuning the language model with specific domain data, the authors designed input templates to supply the LLM with. These input templates, i.e., prompts, are constructed using information about the learner: previous knowledge, interaction history, and targeted styles of explanations. As such, the feedback the student is provided with is even more personalized.

Such an ITS transformation showcased that the integration of RAG and prompt engineering techniques have improved LLM responses to factual, academic, and subject-specific questions quite significantly. While the overall findings showcase the effectiveness of these ITS modifications, the study has focused on the investigation of single-turn, subject-focused, and fact-oriented LLM responses.

2.2.3 Human-like Tutoring

Although LLMs are capable of generating answers to various questions, their behavior does not fully align with the role that the human tutors take. In fact, ChatGPT or other similar systems function primarily as a highly effective knowledge provider, rather than a tutor-like agent (Nye et al., 2023). Nye et al (2023) have defined the characteristics of an effective tutoring process performed by humans. Firstly, human tutors take the initiative and guide the progression and direction of the interaction with the student. When tutors are responding to the student's questions, they do not simply react to the question but purposefully direct the conversation toward targeted learning outcomes. Secondly, a human tutor, instead of providing a clear answer to the student's question, tends to guide the student by continuous questioning to encourage the student's reflection and their own reasoning on the topic. Lastly, human tutors have a lot of pedagogical experience; therefore, they are aware of the most common student misconceptions about the topic and teaching strategies that could be used to target specific student profiles.

Current LLMs, despite their extensive domain knowledge acquired during pretraining, lack the exposure to such pedagogical methods. While the RAG-enhanced LLM-based ITS architectures for student tutoring - especially in knowledge-intensive domains - showcase an improvement in LLM responses, the evaluation of responses in most of the studies is restricted to a single-turn interaction. (Borchers & Shou, 2025; Z. Liu et al., 2025). Jiang & Jiang (2024) have evaluated LLM responses in a multi-turn dialogue setting utilizing prompt-engineering techniques without retrieval-based grounding. However, the study remained limited to a physics-oriented context, and responses were evaluated solely for factual correctness. As such, these studies are limited in providing insights into LLMs' capabilities to respond to questions that are less dependent on facts and more reliant on overall tutor-student interaction.

2.3 LLMs as Second Language Tutors

Unlike factual tutoring, second language learning (L2) relies quite heavily on interaction between a tutor and a student, and the progress in L2 learning cannot rely fully on context-free, rule-based instructions alone. Based on the interactionist perspective, although the second language learners must be able to receive an understandable, targeted input from their tutors for successful L2 acquisition, "modifications to the interactional structure of conversations which take place in the process of negotiating a communication problem help to make input comprehensible to an L2

learner” (Ellis, 1991, p. 4). As such, the ability to actively resolve semantic misunderstandings during the conversation and receive feedback on linguistic errors facilitates their understanding of the language (Loewen & Sato, 2018). More recent work emphasizes the opportunities that language learners have when engaging in conversation. Papi & Hiver (2025) argue that although interaction and corrective feedback are useful for successful L2 learning, they are not sufficient on their own. Rather, what makes interaction beneficial is the learner’s willingness to proactively seek and respond to feedback, ask questions, notice gaps in their knowledge, and use interaction as a way of testing and developing their language abilities. Therefore, the benefit of interaction with a tutor is not only the ability to receive targeted input but also the opportunity to actively participate in the overall learning process. As such, it is necessary to explore whether LLM-based tutoring systems can enable L2 learners with opportunities for active language use.

2.3.1 Intelligent Computer-Assisted Language Learning

While ITS is designed to support the learner across diverse educational domains, tutoring systems for language learning are investigated within a more specialized field. Recently, researchers have started to investigate the field of Intelligent Computer-Assisted Language Learning (ICALL) more increasingly (G. L. Liu et al., 2025). Computer-Assisted Language Learning (CALL) is a discipline that investigates how the development of digital technologies can assist in L2 learning (Chen et al., 2021). The addition of artificial intelligence (AI), together with principles from ITS, has led to the development of ICALL, which enables the system to model the understanding that the student (the learner) possesses, and the language that is being learned (Meurers, 2021). Through the applications of natural language processing (NLP) techniques, ICALL systems can analyse the student’s language production, and subsequently, adapt learning materials and exercises for targeting individual needs of the learner (Ruiz et al., 2023). In fact, the use of NLP enables the ICALL system with a so-called intelligence, which is critical for evaluation and monitoring of the student’s linguistic complexity and overall progress in L2 learning (Meurers, 2021; Ruiz et al., 2023).

As the role and benefits of LLMs within education are already acknowledged, many researchers are examining the capabilities of LLMs within ICALL systems. Lee et al. (2024) have integrated LLM into the ICALL system, designed to support skills of L2 reading. The idea was to focus on the effectiveness of teaching rather than student learning, and therefore facilitate teachers’ day-to-day tasks, such as creating reading questions, grading student assignments, and providing feedback to students using the capabilities of LLM. In the paper by Bodnar (2025), LLM is proposed to take another role - to facilitate the learning of L2 grammar. Here, the author proposes a prototype system that, by utilizing LLM, could rewrite authentic texts to ensure that the grammatical forms the learners are practicing are included in the text. In both cases, the LLM is integrated into a structured tutoring environment, where the teachers are expected to be involved in the supervision of the LLM-generated texts.

ICALL research, however, involves investigating a much wider range of topics, one of which is LLMs as conversational tutoring agents for L2 learning (Panagiotidis, 2024). As such, within these systems, language learners could interact with LLMs directly. However, since the interaction between the LLM and a student is not supervised or monitored by teachers, it is crucial to understand the behaviour of the LLM over an extended course of interaction. Moreover, for L2 learning, it is necessary for LLM to adapt to the student's level of language proficiency; therefore, for each proficiency level, the model must be aware of its pedagogical goals.

2.3.2 Common European Framework of Reference

Proficient User	C2	Can understand with ease virtually everything heard or read. Can summarise information from different spoken and written sources, reconstructing arguments and accounts in a coherent presentation. Can express him/herself spontaneously, very fluently and precisely, differentiating finer shades of meaning even in more complex situations.
	C1	Can understand a wide range of demanding, longer texts, and recognise implicit meaning. Can express him/herself fluently and spontaneously without much obvious searching for expressions. Can use language flexibly and effectively for social, academic and professional purposes. Can produce clear, well-structured, detailed text on complex subjects, showing controlled use of organisational patterns, connectors and cohesive devices.
Independent User	B2	Can understand the main ideas of complex text on both concrete and abstract topics, including technical discussions in his/her field of specialisation. Can interact with a degree of fluency and spontaneity that makes regular interaction with native speakers quite possible without strain for either party. Can produce clear, detailed text on a wide range of subjects and explain a viewpoint on a topical issue giving the advantages and disadvantages of various options.
	B1	Can understand the main points of clear standard input on familiar matters regularly encountered in work, school, leisure, etc. Can deal with most situations likely to arise whilst travelling in an area where the language is spoken. Can produce simple connected text on topics which are familiar or of personal interest. Can describe experiences and events, dreams, hopes and ambitions and briefly give reasons and explanations for opinions and plans.
Basic User	A2	Can understand sentences and frequently used expressions related to areas of most immediate relevance (e.g. very basic personal and family information, shopping, local geography, employment). Can communicate in simple and routine tasks requiring a simple and direct exchange of information on familiar and routine matters. Can describe in simple terms aspects of his/her background, immediate environment and matters in areas of immediate need.
	A1	Can understand and use familiar everyday expressions and very basic phrases aimed at the satisfaction of needs of a concrete type. Can introduce him/herself and others and can ask and answer questions about personal details such as where he/she lives, people he/she knows and things he/she has. Can interact in a simple way provided the other person talks slowly and clearly and is prepared to help.

Table 1: Global scale of Common European Framework of Reference levels. Note. From Common European Framework of Reference for Languages: Learning, Teaching, Assessment (Council of Europe, 2001).

The Council of Europe has introduced a **Common European Framework of Reference (CEFR)**, a framework that defines standardized guidelines for teaching, assessing, and learning the language (Council of Europe, 2001). The purpose of this framework is to describe what learners must achieve to communicate in the most effective way. CEFR defines learners' competences in writing, reading, speaking, and listening across six language proficiency levels, ranging from A1 (the beginner level) to C2 (the advanced level) (Kantarctoglu & Papageorgiou, 2012).

Each level is described in detailed “*can-do*” statements, which are used to monitor the progress of language learners throughout the competence development (Prajapati, 2022). There are three main well-known tables that are typically used for providing an overview of CEFR levels: the global scale (Table 1), the self-assessment grid, and the chart of qualitative aspects of the spoken language use (Council of Europe, 2001). The global scale is a simple overview of each language proficiency level and is designed to clarify the language skills that learners are expected to acquire at a specific CEFR level. The self-assessment grid is a more detailed overview of each CEFR level, describing the competences the learner must acquire across reading, listening, writing, and speaking skills. The aim of such an overview is to allow teachers and language learners to self-assess learners’ current level of language proficiency. Lastly, the table of qualitative aspects of spoken language use provides descriptions of each level based on the features that describe the quality of language production, such as fluency or coherence (Council of Europe, 2001). Since these and many more scales have their own official translations into many languages, they are often utilized as descriptions to define language complexity not only for L2 teaching but also to guide the LLM behaviour.

2.3.3 LLM Behaviour with CEFR as Prompting Strategy

For different text generation tasks, LLMs tend to generate text at a language production complexity level that is usually close to that of a native language speaker, resulting in LLM output that is merely inappropriate for the learners at the beginner level. As such, control of LLM proficiency is key. One of the simplest ways to control LLM behaviour is via prompt engineering (Malik et al., 2024). In fact, CEFR-inspired prompting techniques were widely explored among researchers. Marzona et al. (2025) have utilized CEFR-based prompting to evaluate the extent to which LLMs can rewrite English and Italian texts according to the vocabulary and grammar constraints of the A1 proficiency level and showcased that CEFR-based linguistic restrictions are useful in rephrasing the texts to comply with beginner-level requirements. Lin-Zucker et al. (2025) have also investigated prompt engineering as a technique to adapt LLM behaviour to A1-A2 CEFR levels. Here, authors used Chinese character lists relevant to each CEFR level to generate sentences of appropriate difficulty and showcased the capability of the LLM to adapt to explicit linguistic constraints at least the beginner proficiency levels. Benedetto & Buttery (2025) investigated the LLM abilities for CEFR-targeted text simplification, showcasing that the models can produce outputs that correlate with the respective CEFR proficiency levels. However, even if LLMs appear to be able to approximate proficiency levels, it is relatively unknown how much of the internal CEFR representation the models possess, and how well different CEFR-based prompting strategies work (Benedetto et al., 2025).

Filippetti et al. (2025) took a step further from a simple, single turn exchange between the user and the LLM to a more interactive L2 learning environment. The authors have developed a CEFR-informed chatbot for Italian language learning. Similar to the previous studies, the proposed L2 learning environment was constrained using prompting techniques. During this pilot study, L2

learners were able to practice their language skills in multi-turn, scenario-based interactions, which were found to positively affect overall perceptions of the learning environment, understanding of language, and motivation to learn. However, the study did not involve the analysis of the extent to which the LLM text generation is constrained to CEFR-based levels, and therefore, it is unknown whether the linguistic complexity throughout the conversation is maintained. Importantly, the latter became a center of discussion in a study by Almasi & Kristensen-McLachlan (2025).

2.4 Alignment Drift

Almasi & Kristensen-McLachlan (2025) have simulated multi-turn interactions between a tutor and a student using LLM, with the aim of evaluating the extent to which CEFR-informed (see *Table 1*) prompting is able to constrain the language complexity of the tutor part of the dialogue (generated by a tutor-LLM) in Spanish. A single simulated dialogue between a tutor and a student comprised 9 turns, and although the CEFR-based prompting seemed to be useful in constraining the tutor-LLM behaviour at the start of the conversation, the impact of prompting was found to eventually disappear. The authors have described this phenomenon as an **alignment drift**. The alignment drift has been measured using the structural complexity and readability metrics. While prompting was useful to induce CEFR-level distinctions based on the readability metrics used, the structural complexity metrics - text length and mean dependency distance - failed to show such a clear distinction. Moreover, although some CEFR level distinctions were found, it still appeared as if the tutor-LLM generated text might be too complex for a student at a respective language proficiency level.

The authors have discussed several limitations that could possibly affect the overall findings and conclusion of the study, one of them being the English system prompting for generating output text in Spanish. Such a prompting design could have been the reason why the authors witnessed the English text in the LLM output. Moreover, although the tutor-LLM behaviour has been prompted based on specific CEFR level constraints, the student-LLM has not been constrained at all, resulting in the possibility of the student-LLM output influencing the tutor-LLM responses.

Despite the findings, the change in LLMs' behaviour later in conversation cannot be reliably generalized from the study of a single, widely studied language like Spanish. Moreover, it is necessary to address the limitations that could potentially influence the conclusion of the study and therefore, it is important to extend the investigation and explore whether such effects can be found in other, low-resourced languages.

2.5 Lithuanian as a Case Study

Lithuanian is a perfect example of an **under-explored, low-resource language**, as research on this language within the context of LLMs is relatively limited. Lithuanian is one of the oldest languages that retained many “archaic” linguistic properties (Rusakeviciene & Kriaucionyte, 2012, p. 1),

meaning that it has retained many of its older linguistic features as opposed to other Indo-European languages (Stankevičius & Lukoševičius, 2022). As such, Lithuanian is generally considered to be a morphologically complex language: noun words can take seven different roles (grammatical cases) depending on their function/syntactic role in the text, verbs can take several tense forms and change by number (singular or plural) and person (1st, 2nd and 3rd); verb participles can adapt words to take role of an adjective and match the form of the noun they go with (Ramonienė & Press, 2015). In English, grammatical relations are often expressed through the order and function words such as prepositions (Kennedy, 2003), whereas grammatical relations between Lithuanian words are expressed through word inflections (Stankevičius & Lukoševičius, 2022). The order of words in Lithuanian does not perform the same function as in English. In fact, the word order alters the overall meaning (Rusakeviciene & Kriaucionyte, 2012). Moreover, the complexity of the Lithuanian language further increases due to the ability to form more words by adding prefixes or suffixes to the beginning or the end of the word, respectively. Nevertheless, teaching Lithuanian as L2 is quite challenging, not only due to overall language complexity but also due to the lack of availability of tools that other languages, such as English, have (Valienė, Rasa, 2020).

2.5.1 Lithuanian Pedagogic Corpus

It is only quite recently that Lithuanian began to be taught as L2 for communication purposes. As such, there is very little research on the acquisition of Lithuanian as an L2 (Ramonaite, 2015). In fact, only recently there has been more work ongoing on creating a corpus for teaching and learning the language. One of the recent resources for learning Lithuanian as L2, the **Lithuanian Pedagogic Corpus**, was introduced by Kovalevskaitė & Rimkutė (2020). The corpus itself consists of approximately 670k tokens. This corpus consists of tokens belonging to a written (~619k tokens) and spoken language. The part of the corpus with written language contains 96k tokens that belong to the A1-A2 CEFR proficiency levels, and 523k tokens to B1-B2 levels (Kovalevskaitė et al., 2020). The written language text data for this resource was collected from the Lithuanian coursebooks and authentic materials, with a focus on extracting the text only based on teaching-relevant genres and scenarios. Right now, the Lithuanian Pedagogic Corpus is integrated within the web interface and can be accessed as a second language learning tool, where users are able to look up words or their lemmas and explore their specific usage cases within a sentence (Kovalevskaitė & Rimkutė, 2020; Rimkutė et al., 2022a).

2.5.2 Lithuanian in the Context of LLMs

When it comes to Lithuanian in the context of LLMs, it is generally unknown if, or to what extent, multilingual LLMs can generalize to such an under-explored language. The large datasets that most language models are trained on usually contain a large proportion of English texts. On one hand, the resulting models are great for English text generation tasks; on the other, they are comparably weak at completing the same tasks in other languages (Qin et al., 2025). Even the large public datasets

designed for multilingual language model pretraining display a significant imbalance in language data representation. For example, in the Community OSCAR language corpora, English language accounts for around 2.5 TiB of text, Spanish - approximately 349 GiB, whereas Lithuanian - only 12.7 GiB of text (Brack et al., 2024). It is quite likely that a similar imbalance is present in various other large-scale datasets as well, suggesting that generally, LLMs might struggle to generate error-free text for such a unique and morphologically complex language.

To develop L2 learner communicative competences in Lithuanian, one of the teaching methods that was found to positively affect the language learning experience was the use of a virtual language learning platform (Valienė, Rasa, 2020). As such, an adaptive, AI-powered tutoring system might further support Lithuanian language learning.

2.6 Motivation

There has been quite a lot of research exploring the role of LLMs in education, and many studies have been conducted to explore different methods and evaluate the extent to which LLMs can facilitate learning of L2. As successful language learning requires multi-turn interaction between a teacher and a student, it is necessary to focus on the investigation of systems with conversational tutoring agents for L2 learning. However, development of such systems poses two clear challenges: first, it is necessary to reliably constrain the agent's (LLMs) behavior so that the complexity of the generated text would not exceed the complexity that the L2 learner can understand; second, it is vital to explore if/to what extent LLM is able to adapt to the learners proficiency level throughout the whole course of the interaction with the learner.

Almasi & Kristensen-McLachlan (2025) showcased that although LLM can generally adapt to the respective CEFR level at the start of conversation, behaviour eventually changes. This suggests that prompt engineering might not be considered as a reliable tool to constrain LLM to specific CEFR levels. However, to gain more evidence for such a conclusion, it is necessary to explore whether alignment drift occurs in other languages; therefore, it is not a phenomenon found only in Spanish. Lithuanian is a morphologically rich, low-resource language that is generally considered to be under-explored. To support the development of tools for learning Lithuanian as L2, it is necessary to explore ways in which successful language tutoring can be performed. The aim of this study is to evaluate whether the same phenomenon - alignment drift - occurs in Lithuanian tutoring dialogue, by adopting the experimental design presented by Almasi & Kristensen-McLachlan (2025) and extending it to account for the previously identified limitations. The study will address the following research questions (RQs):

- RQ 1. Is the phenomenon of alignment drift present when simulating tutor-student interactions for Lithuanian language learning?

- RQ 2. To what extent can the output of tutor-LLM be improved if the system prompting is done in Lithuanian as opposed to English?
- RQ 3. To what extent does the output of tutor-LLM improve when the student-LLM is constrained to a specific CEFR language proficiency level?
- RQ 4. If the phenomenon of alignment drift is present in the Lithuanian language, can tutor-LLM alignment be improved by creating an exploratory experimental design informed by the findings from RQs 1-3?

3. Methods

3.1 Code Availability and Reproducibility Requirements

The code to perform data simulation and reproduce the analysis can be found on the GitHub repository¹. The latter was performed using Python (version 3.12.3) and R (version 4.5.3). The interactive UCloud HPC system, managed by the eScience Center (at the University of Southern Denmark), has been used for all data simulation and analysis tasks.

3.2 Selection of LLM

Since the Lithuanian language is still quite an under-explored language in the context of large language models (LLMs), the process of model selection is not so straightforward. The lack of availability of larger-scale evaluations for the Lithuanian language complicates LLM comparison across different model architectures. However, one of the recent studies provides some useful guidance. Kapočiūtė-Dzikiene et al. (2025) have demonstrated that the Gemma2 27B language model performs quite well in Lithuanian text generation tasks, showcasing a lower rate of lexical hallucinations than larger and more computationally expensive LLMs, such as Llama 3.1 70B. The latter suggests that model size alone might not guarantee better text-generation performance in languages that are under-resourced, and that the model architecture and the data the LLM was trained on might be of greater importance.

To further guide the model selection process, results from the EuroEval benchmarking framework were considered. EuroEval² is a comprehensive benchmarking framework for evaluating language models. It offers leaderboards that rank the models based on their performance across a variety of tasks in multiple European languages. According to the Lithuanian leaderboard, one of the top-ranked, open-source, **instruction-tuned** (IT) models was the **Gemma3 27B**³ language model. Given the benchmark performance for the Lithuanian language and previous evidence, **Gemma3** was selected as the model family for the experiments conducted in the present study.

The Gemma3 model family, introduced by Google DeepMind (Kamath et al., 2025), is a family of multimodal, lightweight, open language models. To ensure the methodological comparability with the study conducted by Almasi & Kristensen-McLachlan (2025), the **Gemma3 12B IT** model was included in the experimental setup. Additionally, the current study extended its design by additionally including a smaller variant, **Gemma3 4B IT**, to investigate the influence of the model

¹ GitHub repository: <https://github.com/JustinaRaz/msc-thesis.git>

² EuroEval: <https://euroeval.com/>

³ Model selection and data simulation was performed in January/February 2026. As such, the model selection was based on the top-rated LLMs that were available at the time, with the focus on selecting the open-source model which can be run locally.

scale more systematically. In fact, the Gemma3 language model family is suitable for controlled model comparisons in general: all models can process the same number of tokens within a context window, but each model has been trained on a different amount of data. Specifically, the Gemma3 4B IT (further, *Gemma3 4B*) language model has been trained on 4T tokens, Gemma3 12B IT (further, *Gemma3 12B*) on 12T tokens, and the last, Gemma3 27B IT (further, *Gemma3 27B*) on 14T tokens. As such, variation in model behaviour can be interpreted more clearly: since the context window (i.e., the number of tokens a model can process) remains constant, differences in behaviour can be attributed to other model-specific factors, such as the training data, rather than context size. Most importantly, compared to the older Gemma2 family, Gemma3 has been trained on more multilingual data, suggesting an improved coverage of different languages (Kamath et al., 2025).

3.3 Experimental Design

3.3.1 LLM Inference Setup

As the present study largely follows the study by Almasi & Kristensen-McLachlan (2025), the inference of each LLM was set up in the same way, by using *transformers* package from HuggingFace (Wolf et al., 2020) and setting hyperparameters to the same values to ensure the most comparable model behaviour: *temperature* = 1, in order to allow the model to be creative and produce diverse text simulations; *top_k* = 50, to constrain the model's choice to the top 50 most likely tokens that can appear next; *top_p* = 1, to allow the model to perform the inference by considering all probable words; *min_p* = 0.05, to prevent the model from sampling the tokens that are very unlikely; *repetition_penalty* = 1.1, to reduce the probability for the model to select tokens that have previously been already selected. All three LLMs have been run using two L40 NVIDIA GPUs, with precision set to *bfloat16* for improving the computational efficiency (Yuan et al., 2025).

3.3.2 Simulating a single Tutor-Student Conversation

All conversations between a tutor and a student have been simulated following Almasi & Kristensen-McLachlan (2025) study design as well. Since LLMs are stateless (LLMs generate text as a response to a single user's query), it is necessary to construct an "external memory" - a history of a conversation - which can be supplied to a LLM throughout the whole tutor-student dialogue (Salem et al., 2026). Almasi & Kristensen-McLachlan (2025) achieved it by instantiating an LLM with its own conversation history from a tutor's and a student's perspective and supplying them to an LLM alternately to simulate full tutor-student dialogue. The full dialogue consists of 9 tutor and 9 student turns. As in the study by Almasi & Kristensen-McLachlan (2025), an LLM instance that generates the tutor part of the dialogue will be referred to as tutor-LLM, whereas an LLM instance for student output simulation will be referred to as student-LLM.

3.3.3 System Prompts

3.3.3.1 Reproducibility of Alignment Drift for Lithuanian

A You are an AI language tutor assisting a student in learning Lithuanian by acting as their dialogue partner. The student is learning Lithuanian at C1 level in the Common European Framework of Reference (CEFR) which means that the user is at an advanced level of Lithuanian. As an AI tutor, your instructions are: 1. Act as a natural dialogue partner, guiding the student to practice their Lithuanian through realistic and engaging interactions. This means asking the student questions in a natural manner. 2. Limit your grammar and vocabulary to C1-level Lithuanian (advanced level). 3. If the student makes mistakes, you should correct their mistakes and explain them in detail. 4. Keep everything in Lithuanian. Remember to keep the conversation at a language level appropriate for an advanced learner of Lithuanian. Specifically, the aim for the student is to master the following language skills after C1: CEFR - The student can understand a wide range of demanding, longer texts, and recognise implicit meaning. - The student can express him/herself fluently and spontaneously without much obvious searching for expressions. - The student can use language flexibly and effectively for social, academic and professional purposes. - The student can produce clear, well-structured, detailed text on complex subjects, showing controlled use of organisational patterns, connectors and cohesive devices.	B You are a student learning Lithuanian, responding to a teacher who is facilitating a natural dialogue with you.
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Table 2: System prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at C1 CEFR level. For text generation, the highlighted words within the tutor prompt were changed to represent the A1 and B1 CEFR levels accordingly. The description of the student's language skills (A) is identical to the description of the CEFR Global Scale (Council of Europe, 2001). Both prompts (A and B) are taken from Almasi & Kristensen-McLachlan (2025) study and modified slightly to accommodate the investigation of the Lithuanian language.

RQ 1 refers to a mere phenomenon of alignment drift reproducibility for the Lithuanian language. As in the paper by Almasi & Kristensen-McLachlan (2025), the tutor part of the dialogue was simulated by instructing the model with system prompts that were designed to reflect the student's proficiency at beginner, intermediate and advanced (A1, B1 and C1) CEFR language levels. The system prompt for the latter language proficiency level is showcased in *Table 2A* (system prompts for the A1 and B1 CEFR levels can be found in Appendix A.1). The student part of the dialogue was simulated by providing the LLM with the system prompt showcased in *Table 2B*. Both prompts were modified only to accommodate the Lithuanian language; otherwise, they remain identical to the ones used by Almasi & Kristensen-McLachlan (2025).

3.3.3.2 Lithuanian-Conditioned Prompting for Investigating the Alignment Drift

CEFR	A Tu esi kalbos mokytojas, paremtas dirbtiniu intelektu, kuris padeda mokiniui mokytis lietuvių kalbos dialogo su mokiniu metu. Mokins mokosi lietuvių kalbos C1 lygiu pagal Bendruosius Europos kalbų metmenis (BEKM), tai reiškia, kad jis yra pažengusio lygio mokinsys. Atlikdamas mokytojo vaidmenį, laikykitės šių nurodymų: 1. Bendrauk kaip natūralus dialogo partneris ir padėk mokiniui praktikuoti lietuvių kalbą per realius, įtraukiančius pokalbius. Tai reiškia, kad su mokiniu kalbėsi taip, kaip tai vykėtų įprastame dialoge. 2. Vartok tik C1 lygio lietuvių kalbos gramatiką ir žodyną (pažengęs lygis). 3. Jei mokinsys daro klaidų, jas pataisyk ir aiškiai paaiškink. 4. Viską atlik tik lietuvių kalba. Atmink, kad pokalbis turi atitikti pažengusio lietuvių kalbos mokinio lygį. C1 lygio mokinsys: - Geba suprasti ilgus ir sudėtingus informacinius tekstus. - Geba suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su jo veiklos sritimi. - Geba bendrauti laisvai ir spontaniškai beveik neieškodamas raiškos priemonių. - Geba veiksmingai ir lanksčiai vartoti kalbą socialiniais ir profesiniais tikslais. - Geba tiksliai formuluoti idėjas ir nuomones, sumaniai prisidėti prie pašnekovų nuomonių. - Geba aiškiai, detaliai apibūdinti dalykus, išplėtoti ir apibendrinti pateikdamas tinkamas išvadas. - Geba aiškiai išreikšti savo mintis geros struktūros ir tinkamo ilgio tekstuose.
	B Tu esi studentas, besimokantis lietuvių kalbos, atsakantis mokytojui, kuris padeda palaikyti natūralų dialogą su tavimi.

Table 3: Lithuanian system prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at C1 CEFR level. For text generation, the highlighted words within the tutor prompt were also changed to represent the A1 and B1 CEFR levels accordingly. Lithuanian system prompts for A1 and B1 levels can be found in Appendix A.2.

RQ 2 refers to the investigation of the extent to which the LLM output can be improved when supplying the model with system prompts in Lithuanian, rather than in English. To do this, the part of the system prompts that are not covered by CEFR Global Scale descriptors (as in Table 2) were translated from English to Lithuanian by a native Lithuanian language speaker. Despite being translated into many European languages, the CEFR Global Scale had no official Lithuanian translation available at the time of analysis, and therefore, the CEFR part of the system prompt could not be simply substituted with the official translation. As such, the translation has been taken from a private Lithuanian language school website - Kalba.lt⁴. Table 3 showcases Lithuanian system prompts for generating the tutor and student parts of the dialogue.

⁴ Website: <https://kalba.lt/en/kas-tas-bekm-ir-ka-reiskia-raides-ir-skaiciai-prie-kalbu-kursu/>

3.3.3.3 Constrained System Prompting for Tutor-Student Simulation

A	You are a student who is learning Lithuanian at C1 level in the Common European Framework of Reference (CEFR) which means that you are at an advanced level of Lithuanian. As C1 level student you are able: - To understand long and complex factual and literary texts, appreciating distinctions of style. You can understand specialised articles and longer technical instructions, even when they do not relate to your field. - To express yourself in clear, well structured text, expressing points of view at some length. You can write about complex subjects in a letter, an essay or a report, underlining what you consider to be the salient issues. You can select style appropriate to the reader in mind. Your task - respond to a teacher who is facilitating a natural dialogue with you.
B	Tu esi studentas, besimokantis lietuvių kalbos C1 lygiu pagal Bendruosius Europos kalbų metmenis (BEKM). Tai reiškia, kad esi pažengusio lygio mokinys ir turi gebėti: - Suprasti ilgus ir sudėtingus informacinius ar grožinės literatūros tekstus, suvokti stilių skirtumus. - Suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su tavo veiklos sritimi. - Išreikšti savo mintis aiškiais ir gerai struktūrizuotais tekstaais bei gana išsamiai apibūdinti savo požiūrį. - Rašyti apie sudėtingus dalykus laiškuose, rašiniuose ir ataskaitose, pabrėžti, tavo nuomone, svarbiausius dalykus. - Rašyti įvairius tekstus, atsižvelgdamas (-a) į skaitytoją. Tavo užduotis - atsakyti mokytojui, kuris su tavimi palaiko natūralų dialogą.

Table 4: English (A) and Lithuanian (B) system prompt for constraining the LLM behaviour when simulating students' parts of the dialogue. The highlighted words were changed to represent the A1 and B1 CEFR levels accordingly (see Appendix A.3).

RQ 3 refers to the investigation of the extent to which the LLM output can be improved if the LLM, when simulating the student part of the dialogue, is not only instructed to act as a student, but is constrained to a specific CEFR language proficiency level as well. To mimic the previously described tutor-relevant system prompt across all CEFR levels, a new student system prompt was created using CEFR self-assessment descriptors. In practice, such descriptors are used by the learners themselves to evaluate their own performance within language learning (Piemsai, 2023). The CEFR self-assessment grid describes the learner's own competences in speaking, writing, and understanding the language. Since the interaction with the LLM-based tutor requires the learner to almost fully rely on writing and reading skills, the new system prompt for simulating the student's part of the dialogue included writing- and reading-related CEFR self-assessment descriptors (Table 4). To evaluate the effect of prompting language on the quality of LLM output, prompts in both English and Lithuanian were tested.

It is acknowledged that this prompting strategy might pose a risk for LLM to follow the explicit instructions that might not reflect the language learner's actual behaviour. Nevertheless, it was expected that constraining student's part of the dialogue would ensure more stability in the generated input the LLM receives when generating the tutor's turn, and therefore, enable a clearer evaluation of how well the tutor is able to adapt.

3.3.3.4 Experimental conditions

RQ	Experimental condition	Prompting language	Student-LLM prompt type
1	1	English	Regular
2	2	Lithuanian	Regular
3	3	English	Constrained
	4	Lithuanian	Constrained

Table 5: Overview of the experimental conditions that correspond to the first three RQs.

To address the first three RQs, a set of experimental conditions was defined. Each condition corresponds to a specific configuration of system prompts (*Table 5*). The regular system prompt type used for the student-LLM refers to the student-LLM prompting strategy used in Almasi & Kristensen-McLachlan (2025) study. The constrained system prompt type refers to a new system prompt that instructs the student-LLM to generate text at a specific CEFR language proficiency level, which matches the one provided to the tutor-LLM. The last research question, RQ 4, depends on the RQ 1 - RQ 3 findings; therefore, it is not addressed through a specific experimental condition as the first research questions are. RQ 4 is revisited in section 3.6.

3.3.4 Running the Simulation

The overall simulation of tutor-student conversations was performed following the same design used by Almasi & Kristensen-McLachlan (2025). Each experimental condition - 1 to 3 (see *Table 5*) - was tested on each Gemma3 language model. To capture the overall behaviour, each LLM was set to generate 30 tutor-student conversations per condition and per each type of CEFR language proficiency level (A1, B1 and C1), resulting in 360 conversations simulated by each LLM and 1080 dialogues simulated overall.

Throughout the simulation process, Almasi & Kristensen-McLachlan (2025) used an algorithm to detect non-Spanish language within the generated sentences, and once another language was detected, the LLM was set to re-generate the whole turn. Early in the simulation process, it was found that when simulating text in Lithuanian, all models tended to generate quite a lot of English and sometimes Chinese text:

“Jis vis dar bandydavo tvarkyti namus, valgyti, dirbti – tarsi nieko 发生发生发生发生发...”

“Džiugu girdėti, kad viskas juda pirmyn! Ir tikrai lauksiu naujienų apie tavo prezentaciją. Nesivaržyk kreiptis, jeigu prireiks kokios nors pagalbos! Sėkmės toliau! (I'm happy to hear that everything is moving forward! And I will definitely wait for news about your presentation. Don't hesitate to reach out if you need any help! Best of luck going forward!)"

However, even when the model was re-generating the turn, it would still include quite a lot of non-Lithuanian sentences in the text and fail to simulate clean output even after ten tries. For this reason, the language detection algorithm was removed from the conversation simulation pipeline and included in the text preprocessing pipeline instead.

3.4 Data preprocessing

Before proceeding to data analysis, the simulated dialogues were briefly inspected to see what type of preprocessing is required. During the first preprocessing step, all English-only sentences have been removed from the text. LLMs were found to add quite a lot of emojis (both as icons and textual sequences of symbols) and new-line symbols, such as:

"Labas rytas! Kaip laikaisi šiandien? 😊 (Good morning! How are you doing today?)"

"Taip, norėčiau pabandyti su kitu sakiniu! Pradėkime! :)"

"\n\nNorėčiau daugiau pavyzdžių."

Such noise in the data was removed as well. Subsequently, preprocessing was performed on each word at a time: if the word was longer than the longest known Lithuanian word, it was removed from the sentence; if the word included non-Lithuanian characters, it was removed from the sentence as well. Lastly, if a sequence of word repetition was detected, only the first word in the sequence was kept and the rest was removed. As such, word-level preprocessing enabled the removal of white-space separated and concatenated word repetitions, as well as non-Lithuanian words.

3.5 Data analysis

3.5.1 Reference Data

To evaluate whether the LLM-generated text corresponds to the intended language proficiency level, it is crucial to compare it to CEFR-level-appropriate texts used in L2 learning contexts. The dataset of the previously introduced Lithuanian Pedagogic Corpus (Kovalevskaitė et al., 2020) was not openly available at the time of the analysis. As such, a subset of the corpus was obtained through direct communication with one of the authors (Rimkutė et al., 2022b). The provided sub-corpus consists of texts derived from the Lithuanian coursebooks, consisting of 320 short A1 level texts and 115 longer, B1 level texts. The texts received were already tokenized and included word-level morphological annotations. In total, there are 24,976 A1 level tokens and 23,003 B1 level tokens. According to the author, the morphological annotations were performed automatically and are estimated to have an accuracy of 95%.

The C1 level texts are not a part of the Lithuanian Pedagogic Corpus. However, Bandoriūtė-Leikienė et al. (2023) has published an openly available digital book that contains a diverse selection

of Lithuanian texts that could be used as reference texts for the C1 level. The book itself is intended to be used by adults who are learning Lithuanian as a second language. According to the author, during the time of the book's compilation, texts have been read by A1-C1 level language learners (Bandoriūtė-Leikienė, 2023) to ensure the suitability of these texts. For instance, the learners of the lowest level were assigned to read short and simple poems written for children. While CEFR language proficiency levels are generally considered to be overlapping (Council of Europe, 2020), the distinctions between the higher levels are harder to make. Generally, language learners at A1-B2 levels can be differentiated by the context-dependent language use and the degree to which the use of language is limited. When it comes to the highest levels, C1-C2, the use of language is “unrestricted” (Council of Europe, 2020, p.130) - the language learner should be able to communicate in any context, flexibly, and freely. Since these CEFR levels cannot be described based on specific linguistic features, it makes it difficult to differentiate it from the lower level-relevant texts.

Due to the difficulty of differentiating C1 level language, longer texts from the Bandoriūtė-Leikienė et al. (2023) book were extracted based on the perceived complexity. The amount of text was extracted to approximate the number of tokens within the subsets of A1 and B1 level texts. Approximately three texts were extracted, resulting in a total of 23,525 tokens. Text extraction from the document was performed using the *PyMuPDF* library (PyMuPDF, 2024). To match the A1 and B1 subsets of reference, word-level morphological annotations were produced automatically using the *Stanza* library (Qi et al., 2020). The resulting texts were used as a proxy for C1 level reference data.

3.5.2 Reference Data: Sanity Check

To assess the sanity of the reference data, especially for the C1 language proficiency level, it is necessary to investigate its syntactic, lexical, and morphological features a bit more. Previously, Boizou et al. (2020) delved into the evaluation of a subset of the Lithuanian Pedagogic Corpus. As the whole corpus has been assigned with morphological annotations automatically, the authors' aim was to assess how and which linguistic features correlate with the complexity of the text across the A1-B2 CEFR levels. As such, the reference data was compared against the previously reported sub-corpus-relevant correlates to verify that it exhibits the expected linguistic patterns, and that text complexity increases progressively from A1 to C1 level.

The computation of different parts of speech (POS) frequencies was not specified in the work by Boizou et al. (2020). Moreover, it is not documented which morphological features have been used to extract the overall distributions. As such, the documentation of Universal Dependencies for the Lithuanian language⁵ has been used to define rules for feature extraction (de Marneffe et al., 2021).

⁵ Universal Dependencies for Lithuanian: <https://universaldependencies.org/lt/>

Appendix A.4 contains the overview of how that has been done. *Table 6* provides the frequencies of linguistic features in the reference data.

In terms of morphological features, Boizou et al. (2020) found that finite verb forms are more frequent at lower CEFR levels and less frequent at higher ones. Reference data generally reflects this distribution, with the frequency of finite verbs decreasing from A1 to C1 levels. The overall frequency of participles, adverbial, and half-participles has also been found to correlate with text complexity (Boizou et al., 2020). The progressive increase of these linguistic features is also depicted in the reference data. When it comes to the voice of participles, active participles are considered as more complex than passive ones, suggesting that lower-level texts should contain fewer active and more passive participles (Boizou et al., 2020). The latter is true for the reference data; moreover, active participles become progressively more frequent the higher the level is. However, the opposite is found in frequencies of passive participles - these linguistic features were found to increase, rather than decrease, in frequency as well. Dative and instrumental forms of nouns were also found to depict the text complexity in the Linguistic Pedagogic Corpus (Boizou et al., 2020). In the reference data, these noun forms seem to increase in frequency from A1 to C1 level as well. Lastly, Boizou et al. (2020) suggested that the degree of adverbs and adjectives might help in the evaluation of text complexity, too - positive adjectives/adverbs were found to be more frequent in lower-level texts, whereas comparative adjectives/adverbs were found to be more frequent in higher-level texts. In the reference data, the overall trend for comparative adjectives/adverbs holds, whereas the trend across CEFR levels for positive adjectives/adverbs is not depicted.

In terms of syntactic and lexical features, an increase in average sentence and word length, and the type/token ratio from lower to higher levels were also found to depict text complexity (Boizou et al., 2020). Reference data showcases this trend for average sentence length quite clearly. While average word length and type/token ratio generally depicts this trend as well, frequencies are slightly higher for B1 level texts compared to C1 level texts. Overall, although a monotonic increase or decrease is not found across CEFR levels, the general pattern seems to align with Boizou et al. (2020) reported findings.

It is acknowledged that the performed comparisons do not confirm or validate the reference data as a dataset that depicts the attributes of CEFR language proficiency levels accurately, and therefore, the development of such a dataset is out of the scope of the present study. However, evaluating the reference dataset based on the previously identified measures that correlate with text complexity provides a way to assess whether the texts exhibit the expected linguistic patterns across CEFR levels. Since there are no officially established benchmarks, the dataset was used as a baseline against the LLM-simulated texts.

Morphological features						
CEFR level	Verb forms (finite and non-finite)			Voice and tense of participles		
	Finite	Participles	Adverbial participles	Half participles	Active present	Passive present
A1	159.0 80.8%	5.4 2.8%	0.4 0.2%	0.1 0.0%	0.1 0.1%	2.6 1.3%
B1	130.5 67.6%	25.8 13.4%	3.7 1.9%	2.1 1.1%	4.2 2.2%	7.9 4.1%
C1	124.9 60.2%	42.8 20.6%	12.8 6.2%	2.6 1.3%	5.1 2.5%	8.2 3.9%

CEFR level	Nouns			Degrees of adjectives and adverbs		
	Dative forms	Instrumental forms	Positive - adjective	Comparative - adjective	Positive - adverb	Comparative - adverb
A1	4.5 1.6%	10.3 3.8%	52.1 88.3%	0.7 1.1%	107.3 97.5%	2.0 1.8%
B1	9.2 2.7%	22.7 6.8%	60.2 83.9%	2.1 2.9%	70.1 88.8%	5.8 7.3%
C1	10.3 3.8%	26.1 9.6%	51.5 90.1%	4.5 7.9%	69.9 91.6%	5.2 6.9%

Lexical and syntactic surface features			
CEFR level	Average sentence length (words)	Average word length	Type/token ratio
A1	4.38	5.41	0.257
B1	10.4	5.95	0.422
C1	13.05	5.64	0.393

Table 6: Frequencies of POS (per 1,000 words) in the reference dataset.
 Frequencies exclude the punctuations. Relative frequencies within the specific grammatical categories are depicted as percentages.

3.5.3 Computing Structural Complexity

To evaluate the effect of system prompts on the simulated tutor part of the dialogue, Almasi & Kristensen-McLachlan (2025) computed Spanish-specific metrics of readability to account for language-specific features. Additionally, authors computed language-generic structural metrics that depict the overall structural complexity of the text. Since there are no known Lithuanian-specific metrics, the present study relied on the features of structural complexity - the Mean Dependency Distance (MDD) and the average length of text.

MDD is generally considered to be a metric representing the syntactic complexity (Ouyang et al., 2022) and can be used as a proxy for evaluating the comprehension difficulty of the text (Haitao Liu, 2008). In simple terms, dependency distance is the distance (in words) between one word, i.e., *the governor*, and another word that complements it, i.e., *the dependent* (Haitao Liu, 2008). Within the research of second language learning, MDD has been shown to be useful for differentiating the text at different language proficiency levels (Ouyang et al., 2022) - higher values of MDD signal greater difficulty of text comprehension. As for the text length, at A1 level learners are described as being able to understand basic and simple phrases or text, whereas at higher levels learners must understand longer (more complex) texts (Council of Europe, 2020), meaning that the higher CEFR level is, the longer the texts should be observed.

The two measures for evaluation of text complexity have been computed in Python using the *TextDescriptives* package (Hansen et al., 2023) together with a medium-sized *spaCy* (Vasilev, 2020) model for Lithuanian (`lt_core_news_md`, version 3.8.0).

3.5.4 Linear Mixed-Effects analysis

The influence of system prompts on LLM-generated output was evaluated by performing an analysis of linear mixed-effects. The analysis was performed by running the model as in Almasi & Kristensen-McLachlan (2025) study, namely:

$$\text{metric}_{\text{model}} \sim \text{level} + (1|\text{chat}_{\text{id}})$$

The dependent variable *metric_{model}* is the model-specific metric derived from the LLM-generated output, which in this case is the MDD and average text length. *Level* refers to the CEFR levels, namely, A1, B1, and C1, defined as a fixed effect. The *chat_{id}* is set as a random effect, which allows it to capture the variability between the chats that were generated.

Initially, for reproducibility reasons, the linear mixed-effects model was run only on those conversations that were generated by supplying the tutor-LLM with a system prompt in English, and the student-LLM with a so-called regular system prompt. As the present study involved a broader investigation, the exact model was run on the conversations that were generated by

providing the tutor and student-LLM with system prompts as defined in *Table 5*. As in the previous study, the Bonferroni correction was used to adjust p-values for multiple comparisons.

3.5.5 Evaluating LLM Outputs against the Reference Data

Additional evaluation of the LLM-generated dialogue outputs against the reference data was necessary to investigate text differences/similarities at specific CEFR language proficiency levels. Compared to the reference dataset, the overall format of LLM-generated text is different. As such, the comparison was performed by combining dialogue-based tutor-specific simulated LLM outputs into continuous texts. The aggregation of different conversations/dialogues was based on the experimental conditions. For instance, all dialogues generated using an English system prompt, in combination with a constrained system prompt used for simulating the student role were concatenated into a single continuous text. Such concatenation enabled not only the estimation of linguistic feature frequencies at specific CEFR levels, but also a way to compare it to the estimates found in the reference data.

The continuous LLM-generated text was processed in the same way as C1 level reference texts - morphological annotations were produced using the *Stanza* library (Qi et al., 2020). To assess the extent to which text complexity in the generated texts matches the complexity of the reference texts, features that were found to differentiate CEFR levels in the reference data the most were extracted from simulated texts (*Table 6*), namely, the average sentence length, dative and instrumental noun forms, and finite and non-finite verb forms. Moreover, since Almasi & Kristensen-McLachlan (2025) computed MDDs for LLM-generated texts, this additional metric was computed for the reference texts as well to enable a direct comparison.

As previously mentioned, the reference texts and LLM-generated dialogues differ in their format (e.g., structure and overall function). The reference dataset was constructed from texts that were originally designed as reading comprehension materials for language learners. The dialogue-based simulations, on the other hand, are supposed to reflect the interaction between the tutor and a student learning Lithuanian as L2. While engaging in a dialogue with the LLM as a tutor requires competence in reading comprehension, such a type of interaction involves skills of written interaction as well (Council of Europe, 2020). Within the CEFR framework, “in written interaction the language used is similar to oral language” and accurately structured language production is not prioritized (Council of Europe, 2020, p. 81). As such, it is quite probable that dialogue-based interactions are often less complex than the aggregated texts for reading. Nevertheless, the two different types of datasets have been compared to one another as there is no other baseline to evaluate the LLM-generated text against. The latter being acknowledged, it is necessary to interpret the comparison results with caution.

3.5.6 Alignment Drift Versus Mutual Adaptation

The phenomenon of alignment drift was described as the deviation in text complexity at a specific CEFR proficiency level over the course of the dialogue (Almasi & Kristensen-McLachlan, 2025). However, the interaction between the two LLMs might involve more complex mechanisms, potentially as complex as the ones within the interaction between humans. In human-to-human interactions, the positive outcome of the conversation is often attributed to the individuals converging to a common ground (Guydish & Fox Tree, 2021). Common ground in a conversation can be defined as a mutual goal between two or more interlocutors, namely, reaching an understanding of one another, which is often facilitated by interlocutors' alignment in similar language use, turn-to-turn behaviours, etc. (Guydish & Fox Tree, 2021).

Similarly to how humans adapt to one another in a conversation, language models appear to showcase a somewhat adaptation as well. Kandra et al. (2025) have shown that within the LLM-to-LLM conversations, the syntax of the simulated texts becomes relatively similar as the interaction progresses. As such, while the alignment drift could be an actual observed phenomenon, the convergence in structural feature values might as well be an indicator of tutor and student-LLMs establishing this so-called common ground (i.e., mutual adaptability).

To identify which of the two explanations is at play, it was necessary to compare the structural metric values extracted from the tutor part of the dialogue against those of the student. As such, metrics of average text length and MDD were extracted from the student-LLM outputs after preprocessing them in the exact way as the tutor-LLM texts (see sections 3.4 and 3.5.3). The difference between the tutor and student outputs was evaluated by computing the Euclidean distance, i.e., the absolute difference between the corresponding text complexity values, where f refers to the specific feature of structural complexity at turn t :

$$\text{Distance}_f(t) = |f_{\text{tutor-LLM}}(t) - f_{\text{student-LLM}}(t)|$$

3.6 Mitigating the Alignment Drift

RQ 4 refers to the exploration of whether the alignment in the tutor part of the conversation can be improved to reflect the CEFR language proficiency level throughout the whole dialogue. The exploratory experimental design is dependent on the findings of RQs 1-3. As such, a new design could not be introduced in detail without the initial findings. Therefore, the general procedure for developing the new experimental design is informed by the findings of Almasi & Kristensen-McLachlan (2025), which provide a methodological basis for adapting the design to the present study after initial results are obtained. The specifics of the experimental design based on the findings of the current study are presented in section 4.2.4.

Based on the performance of Gemma3 12B model in the Almasi & Kristensen-McLachlan (2025) study, the use of CEFR-based system-prompting led to a clear outcome difference across CEFR

levels at the tutor's 3rd and 4th turns. At the 5th turn and onwards, the tutor-LLM showed a tendency to drift - text length became shorter and syntactic complexity decreased, leading to a convergence of values, especially at B1 and C1 levels. Laban et al. (2025) and Z. Liu (2026) state that LLMs generally struggle to perform well in multi-turn interactions. Z. Liu (2026) found that language models tend to disregard the information that is provided early in the context and prioritize the information that was provided more recently. The author has described this effect as a recency bias. Laban et al. (2025) shown that if the LLM generates a sub-optimal turn, it may potentially bias its future responses due to misaligned response re-appearing in the context. As such, if the tutor-LLM produced the output (at 5th turn) deviating from the expected text (generated response at 3rd and 4th turns) at a specific CEFR level, it is quite possible that its recent, and therefore, subsequent turns result in the drift of alignment. These findings imply that the observed alignment drift could be related to the diminishing influence of system prompt on the model's behaviour as the context of the conversation is growing.

Madaan et al. (2023) introduced a simple approach that appears to improve the quality of the LLM-generated output: a framework of self-refinement. In short, the self-refinement framework has three steps: first, the LLM is provided with an input prompt, which it uses to generate the text; second, the LLM is provided with its own generated text and is instructed to provide feedback on it; third, the LLM is instructed to refine its original response by adapting it based on the generated feedback. The language model is expected to perform the refinement until the text meets the pre-specified criteria. Motivated by the findings, the present study introduces self-refinement as part of an exploratory experimental design aimed at mitigating alignment drift in the tutor-LLM responses. Since the approach of self-refinement can improve the quality of generated text, it is hypothesized that including this step in the simulation process would make tutor-LLM responses more aligned. As such, if the self-refinement approach proves to be effective, subsequent tutor-LLM turns should exhibit less deviation.

To ensure more control over the feedback that the tutor-LLM receives as an instruction for text refinement, a controlled, conversation-specific feedback mechanism is proposed. More specifically, feedback was determined based on the metric values used to infer the alignment drift - the average text length and mean dependency distance. Given the Almasi & Kristensen-McLachlan (2025) findings, the means and standard deviations of text length and MDD estimates would be extracted from those tutor-LLM responses that showed the clearest CEFR-aligned behaviour, namely, 3rd and 4th turns. Extracted values would be used to define reference thresholds/boundaries against which the subsequent tutor-LLM responses are evaluated. If the extracted metrics do not fall within the conversation-specific thresholds, the tutor-LLM would receive feedback containing instructions for refinement. *Table 7* shows the corrective refinement prompts. The choice of system-prompting and feedback language is informed by the initial evaluation of the results (results to answer RQs 1-3). The setup of self-refinement is depicted in *Figure 1*.

Mean Dependency Distance	
[Lithuanian version] Paskutinis atsakymas yra struktūriškai per paprastas mokiniui, kuris mokosi lietuvių kalbos B1 lygiu. Naudok šiek tiek sudėtingesnes sakinių struktūras ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too simple for a student who is learning Lithuanian at B1 level. Use slightly more complex sentence structures while keeping the meaning.	[Lithuanian version] Paskutinis atsakymas yra struktūriškai per sudėtingas mokiniui, kuris mokosi lietuvių kalbos B1 lygiu. Supaprastink sakinių struktūrą ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too complex for a student who is learning Lithuanian at B1 level. Simplify the sentence structure while keeping the meaning.
Text Length	
[Lithuanian version] Paskutinis atsakymas yra per trypas mokiniui, kuris mokosi lietuvių kalbos B1 lygiu. Pratęsk atsakymą ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too short for a student who is learning Lithuanian at B1 level. Extend the previous answer while keeping the meaning.	[Lithuanian version] Paskutinis atsakymas yra per ilgas mokiniui, kuris mokosi lietuvių kalbos B1 lygiu. Sutrumpink atsakymą, pašalindamas perteklinę informaciją ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too long for a student who is learning Lithuanian at B1 level. Make the previous answer more concise while keeping the meaning.

Table 7: Feedback prompts used for the exploratory experimental approach of self-refinement. Left-side instructions are provided to the LLM when the respective metric falls below the computed reference thresholds. Right-side instructions are provided when metrics fall above the thresholds. The highlighted words were changed to correspond the C1 CEFR level accordingly (see Appendix A.6).

As illustrated in Figure 1, tutor-LLM responses that show the most stable conversation-specific alignment in structural complexity metrics (e.g., turn n and turn $n+1$) are selected as references for subsequent turns. Text of these responses is pre-processed (according to section 3.4). The latter is required to ensure the accurate number of tokens/words in a text, based on which the thresholds are derived. Each time the tutor-LLM is requested to generate a subsequent response, the simulated text is passed through the pre-processing pipeline, after which structural metrics are computed and compared against the previously derived thresholds. In case the metrics are found to fall within the derived thresholds, the tutor-LLM's response is appended to the conversation history for subsequent generation of the student's response. Otherwise, the temporary conversation history, together with the latest tutor-LLM response and an appropriate refinement instruction, is used to refine the turn. The refined response is passed through the pre-processing and metric extraction steps again, followed by another round of checks.

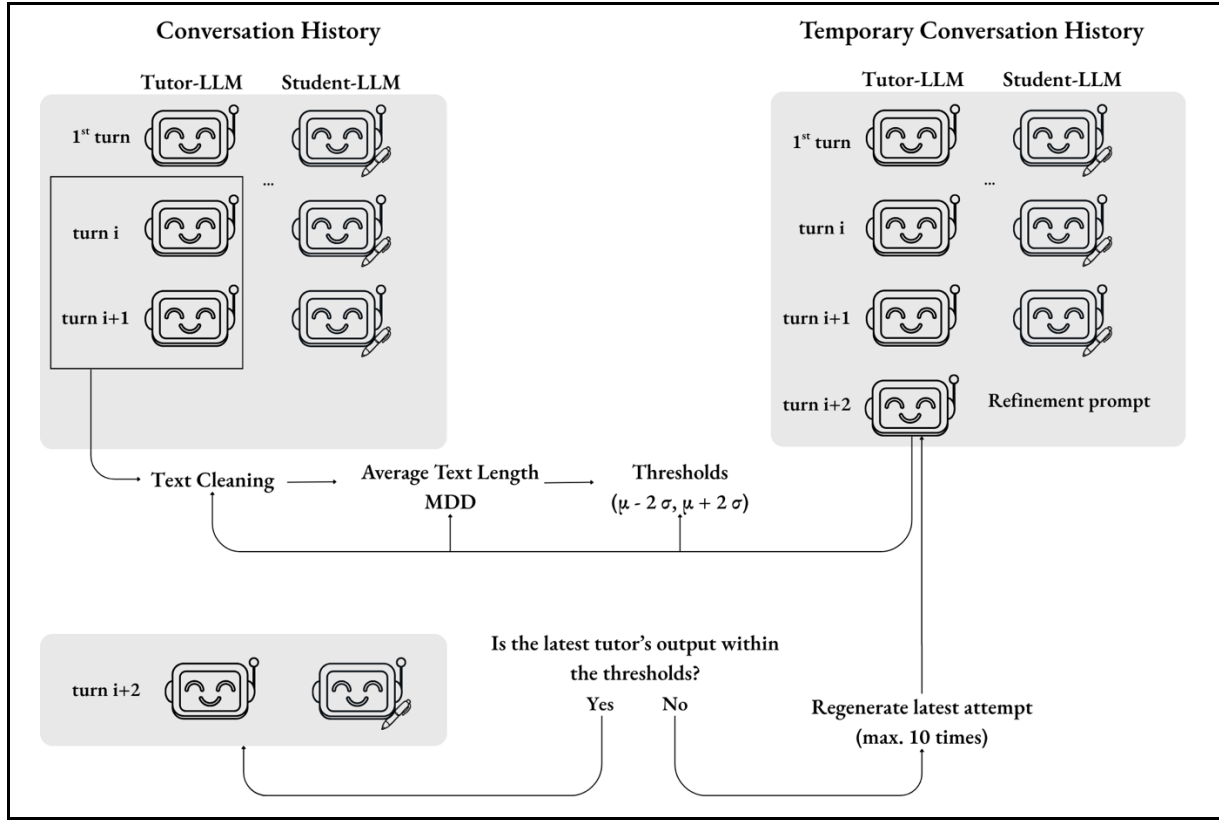


Figure 1: An exploratory experimental setup to mitigate the alignment drift. Turn i and $i+1$ denote CEFR-aligned tutor's responses that are selected as reference turns. The number of reference responses can vary.

The refinement is set to occur for as long as the extracted response-specific metrics fall between the computed thresholds, or as long as the total of 10 attempts is reached. The process is repeated for the rest of the turns until the whole conversation is simulated. As simulation of dialogues is computationally expensive and addressing the RQ 4 requires a new, exploratory experimental design, the testing of the new setup is performed by instructing the selected model to only generate conversations at B1 and C1 level. The number of tutor-student conversations to simulate is kept unchanged. The choice of language for system-prompting and the type of system prompt used to instruct the student-LLM is based on the findings presented in the upcoming section 4.2.4.

4. Results

4.1 Qualitative Findings

4.1.1 Data Preprocessing

The preprocessing of the tutor-LLM-generated responses revealed how different system prompts (Table 5) influence the quality of the LLM output. Figure 2 presents the total number of English sentences, as well as the number of invalid, long, and repetitive words removed from the LLM-generated texts across all four conditions and for each Gemma3 model.

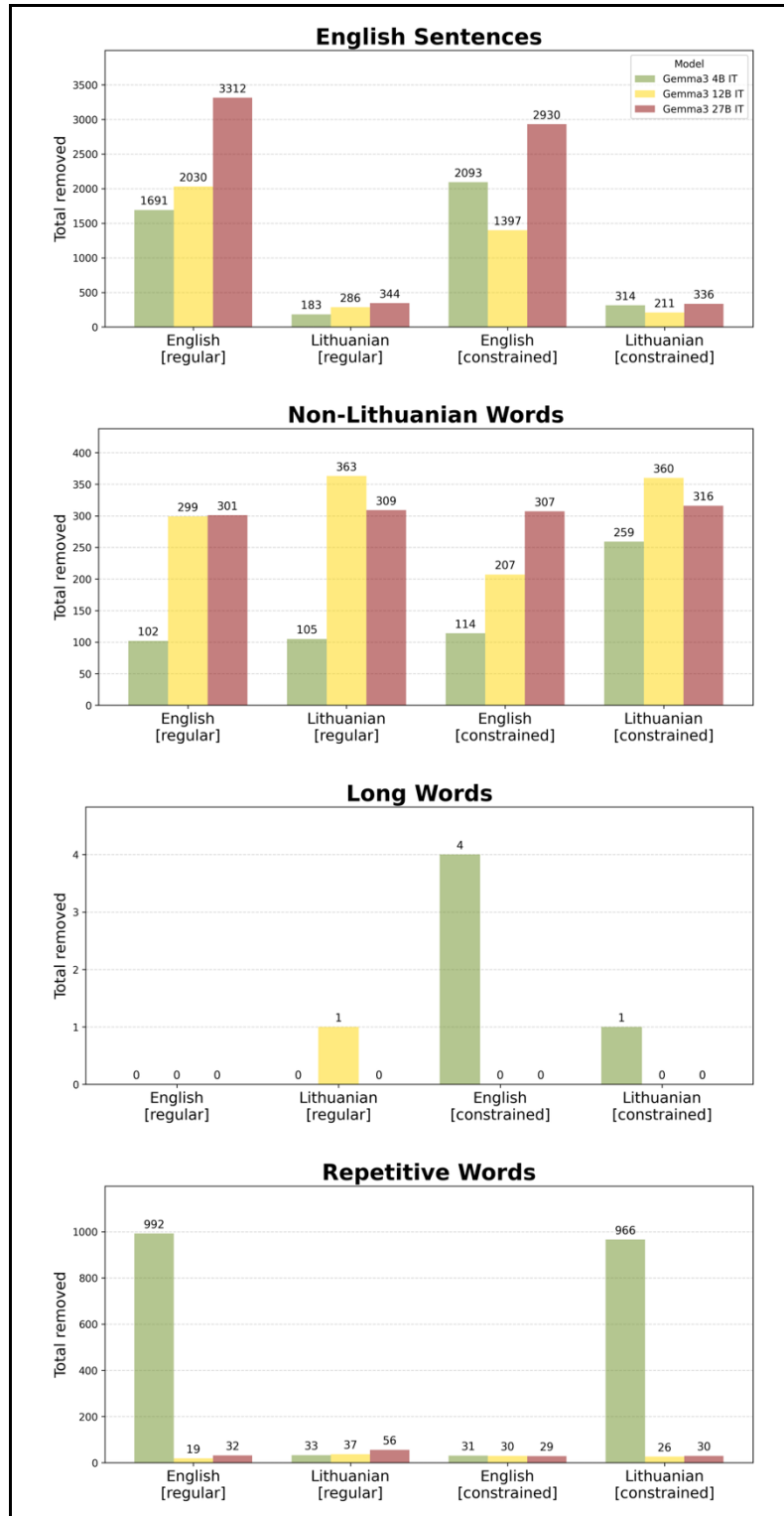


Figure 2: Errors in LLM-generated output across the conditions. The figure depicts the total number of instances removed from raw LLM-as-tutor text. Non-Lithuanian words are words generated using characters that do not belong to the Lithuanian alphabet (e.g., Chinese). Long words refer to words that contain more characters than the longest Lithuanian word⁶. Repetitive words are single-word duplications within the text.

⁶ Based on the following source: <https://artsphere.org/interactive-programs/world-passport/europe/lithuania/>

For instance, the Gemma3 27B model was found to generate English text more often than the other models across all conditions. In almost all instances, the prevalence of English text within a single response was not a random switch from Lithuanian to English in the middle of the sentence, but rather a translation-like text, such as:

“Puikiai paaškinai! (You explained perfectly!)”

When considering English versus Lithuanian system-prompting, it is clear that instructing all three models in Lithuanian resulted in fewer English sentences within the generated output.

Larger Gemma3 models were found to produce more non-Lithuanian words than the Gemma3 4B language model. However, the latter model was found to experience stability issues more often than the rest of the models: Gemma3 4B language model showed a tendency to generate either a long string of character sequences without spaces, or sequences of repetitive words separated by spaces.

4.1.2 Lexical Validation

During the initial inspection of LLM-generated data, it appeared that all LLMs tended to produce a number of spelling errors, i.e., lexical validity errors. As such, the following investigation involved the evaluation of error rate (lexically invalid word rate) in tutor’s responses across the dialogues and at each CEFR level. The average error rate is depicted in *Figure 3*.

Based on the results, the Gemma3 4B model was found to produce the most lexically invalid words across all conditions and at all CEFR levels compared to other LLMs. Interestingly, the error rate for the latter model was found to be quite high at the A1 level, with a lower error rate observed in the B1 level texts. At the C1 level, the error rate seems to further decrease under conditions where the student-LLM was provided with a constraining system prompt. Otherwise, the error rate, on average, tended to slightly increase.

Larger LLMs, namely, the Gemma3 12B and 27B, generally generated text at a relatively similar error rate. When prompting was performed in Lithuanian, the average error rate across these models was found to largely overlap. However, when the LLMs were prompted in English, the Gemma3 27B model produced slightly fewer lexical errors in B1 level texts compared to Gemma3 12B. At A1 level, however, it produced more lexical errors when the conversational partner, the student-LLM, was constrained to the A1 level as well. Overall, supplying the models with Lithuanian system prompts led to fewer lexical violations, whereas constraining the student-LLM appeared to increase the variation of average error rates across the dialogues.

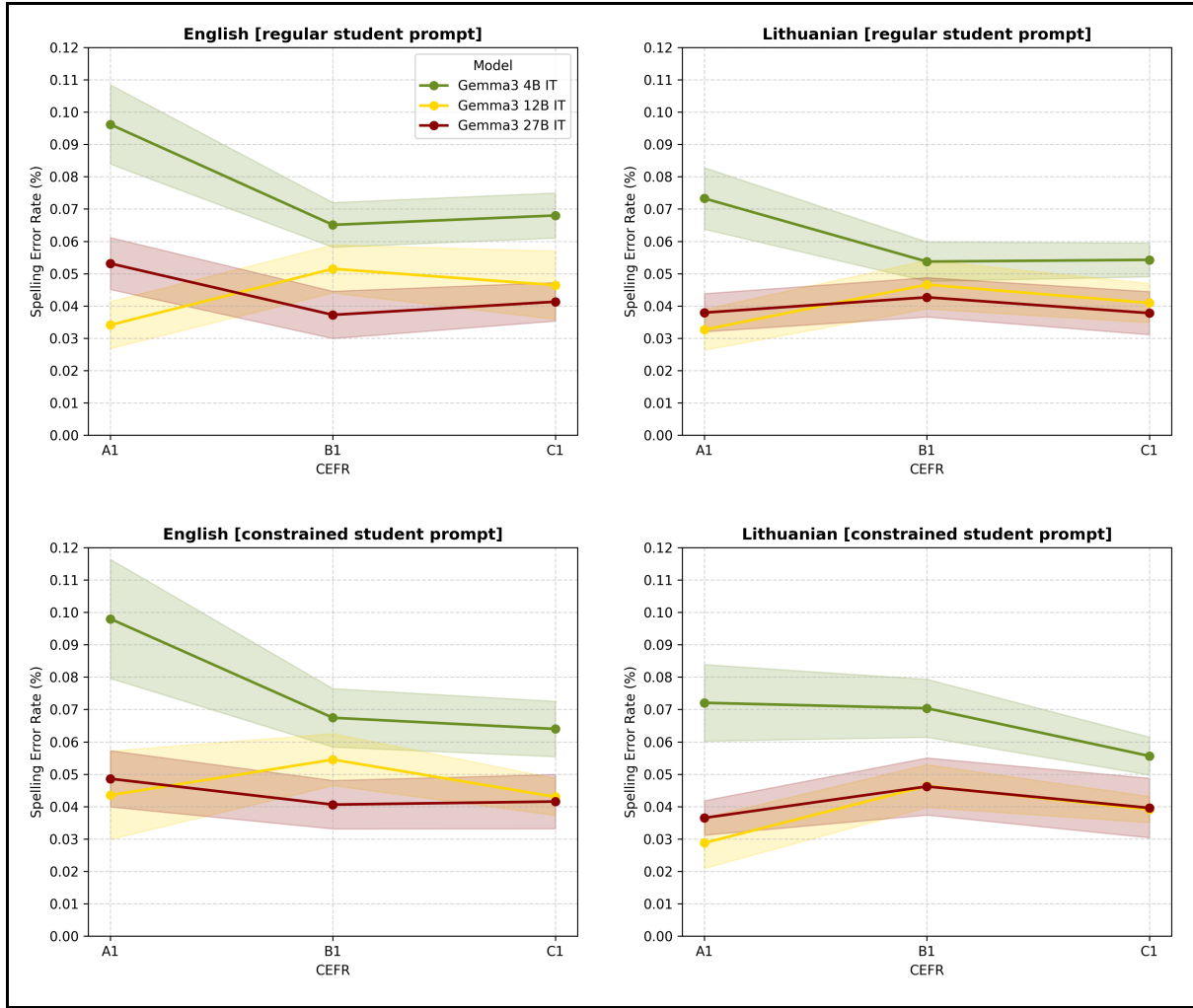


Figure 3: Average dialogue error rate at A1, B1, and C1 language proficiency levels. Each subplot refers to the experimental condition. Shaded areas represent 95% CIs. The Y-axis represents the percentages.

4.2 Quantitative Findings

4.2.1 Alignment Drift

Figures 4-7 showcase the average text length and MDD estimates for each model across all experimental conditions. The findings presented in Figure 4 address RQ 1 by showcasing how the experimental setup defined by Almasi & Kristensen-McLachlan (2025) influenced tutor-LLM responses when the model was prompted to generate Lithuanian text. Overall, there is a clear distinction between the CEFR levels over the time of the dialogue. The general trend appears as expected: average text length and MDD estimates were estimated to be the highest at the C1 level, with B1 level average metrics being slightly lower and A1 metrics being the lowest. The linear mixed-effects analysis complements these overall findings: the average estimates of structural features for intermediate and advanced CEFR levels were found to be significantly higher ($p < 0.001$) than the average estimates for beginner level output (see Appendix A.5.1 for the full estimate table).

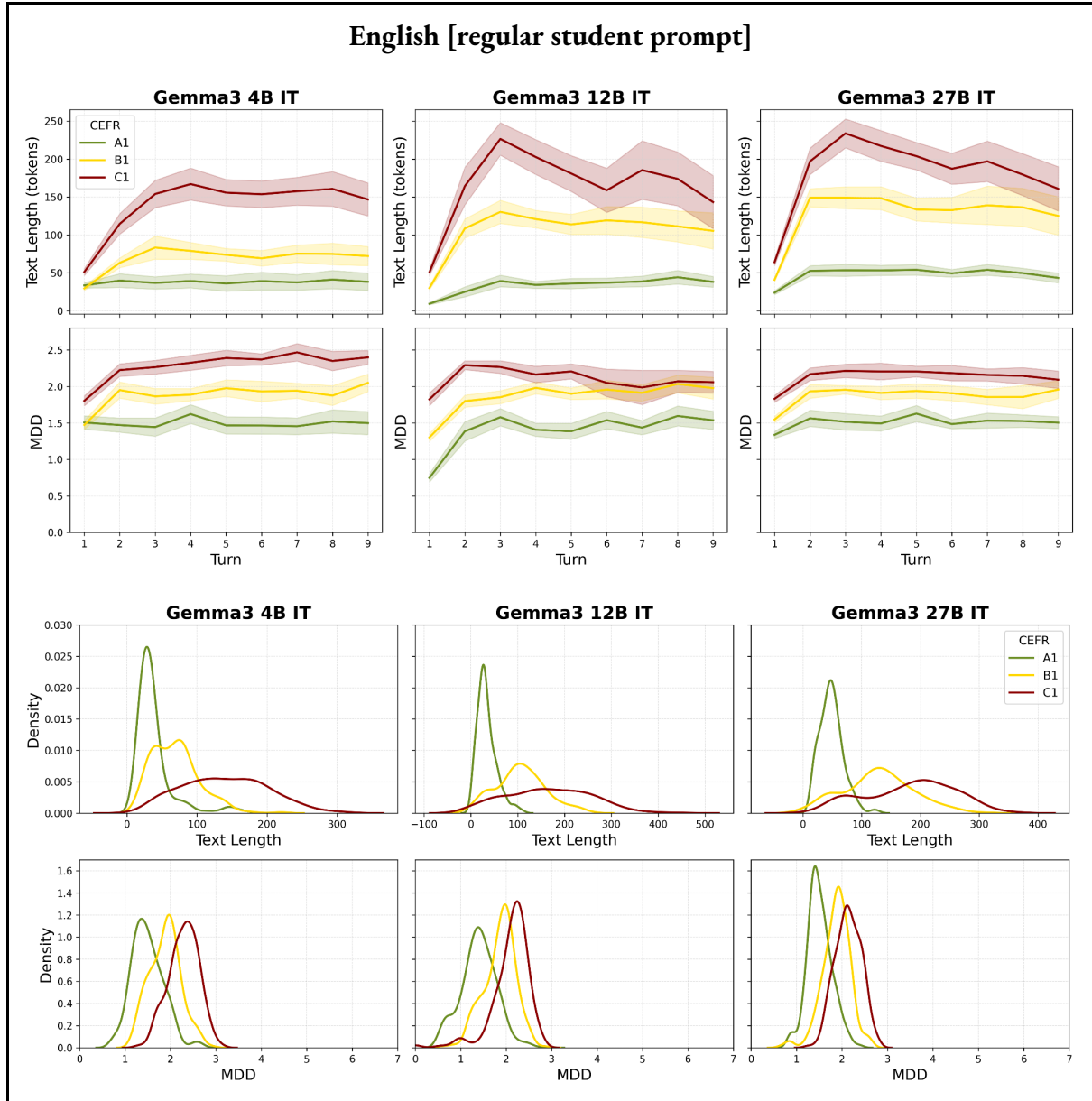


Figure 4: Average text length (tokens) and MDD (mean dependency distance) features extracted from the tutor part of the dialogue when providing the LLM with a system prompt in English using a regular system prompt to simulate the student part of the conversation. The top subplot depicts average values at different time points of the dialogue (across tutor turns). Shaded areas represent 95% CIs. Bottom subplots represent metric densities⁷ - the estimated distributions of each metric value across CEFR language proficiency levels. Note: density plots contain different scales.

The average structural complexity metrics reveal an emerging trend: at the start of the conversation, the complexity of the tutor-LLM responses increases from turn 1 to 2, without any overlap across CEFR levels, eventually reaching a rough stabilization between turns 2 to 4. However, in most cases, towards the end of the conversation, the complexity metrics seem to show a tendency to converge, especially at B1 and C1 CEFR levels for Gemma3 12B and 27B models. The latter is depicted as an

⁷ The x-axis scale in some density plots (also in subsequent figures) is negative as the Kernel Density Estimation (KDE) is used.

overlap in 95% confidence intervals (CIs). As such, although less severely, the phenomenon of alignment drift (Almasi & Kristensen-McLachlan, 2025) seems to appear in Lithuanian conversations as well.

Density distributions showcase estimated metric value distributions. Overall, the text length (in tokens) distributions at B1 and C1 levels for all models appear distinct from those estimated at the A1 level, hence the significant linear mixed-effects findings. Distributions depicting text length for B1 and C1 levels seem to largely overlap, implying that text length at the B1 and C1 levels for larger Gemma3 models is quite similar. In fact, the average estimates from the linear mixed-effects analysis complement this finding: at the A1 level (intercept), the values across the language models are estimated to be quite similar already (*Figure 9*). At the B1 level, Gemma3 12B and 27B models produce responses that are 73 and 80 tokens longer than responses at the A1 level, respectively (*Figure 9*). At the C1 level, the average increases by 132 and 134 tokens for the 12B and 27B models, respectively. The distributions of A1 level output across all models are narrower than those at higher levels, implying greater model stability when generating beginner-level responses. The MDD metric distributions appear to be relatively similar across the models - the distributions across the levels are largely overlapping. The Gemma3 27B model has the least variability in syntactic complexity of the output (the narrowest distributions), with responses estimated to be slightly simpler than those generated by smaller Gemma3 models.

The findings in *Figure 5* address RQ 2 by showcasing Lithuanian system-prompting influences tutor-LLM responses. Based on the average values, the distinction between CEFR levels is still clear. The linear mixed-effects analysis once again reveals that the average estimates at B1 and C1 levels are significantly ($p < 0.001$) higher than estimates at the A1 level (see *Appendix A.5.2*). However, as depicted in the plot, there appears to be even more convergence in average text length values for Gemma3 12B and Gemma3 27B models at B1 and C1 levels. This is particularly evident in the average MDD estimates for Gemma 3 27B, where the B1 and C1 values overlap across almost every turn, as indicated by their 95% CIs. Gemma 3 27B does not show notable alignment drift in text length at the C1 level, as estimated metric values are lower from the start of the conversation compared to those values when system-prompting was performed in English (*Figure 4*). In contrast, alignment drift is still evident for Gemma 3 12B responses, including those at A1 level. A similar pattern is also observed in Gemma 3 4B responses at the A1 level; interestingly, toward the end of the conversation, the average MDD appears to increase. Overall, compared to English system-prompting, alignment drift appears to be more evident when prompting is performed in Lithuanian.

Density distributions for text length are quite wide across all models, at all levels, indicating a great variability in text length across conversations. In this condition, the smallest Gemma3 model produced the least variable text (at A1 level, the maximum text length is estimated to be below 200 tokens, at B1 and C1 levels - below 400 tokens), whereas the output of larger models seems to be more diverse and longer. Density distributions for MDD showcase slightly different findings. All

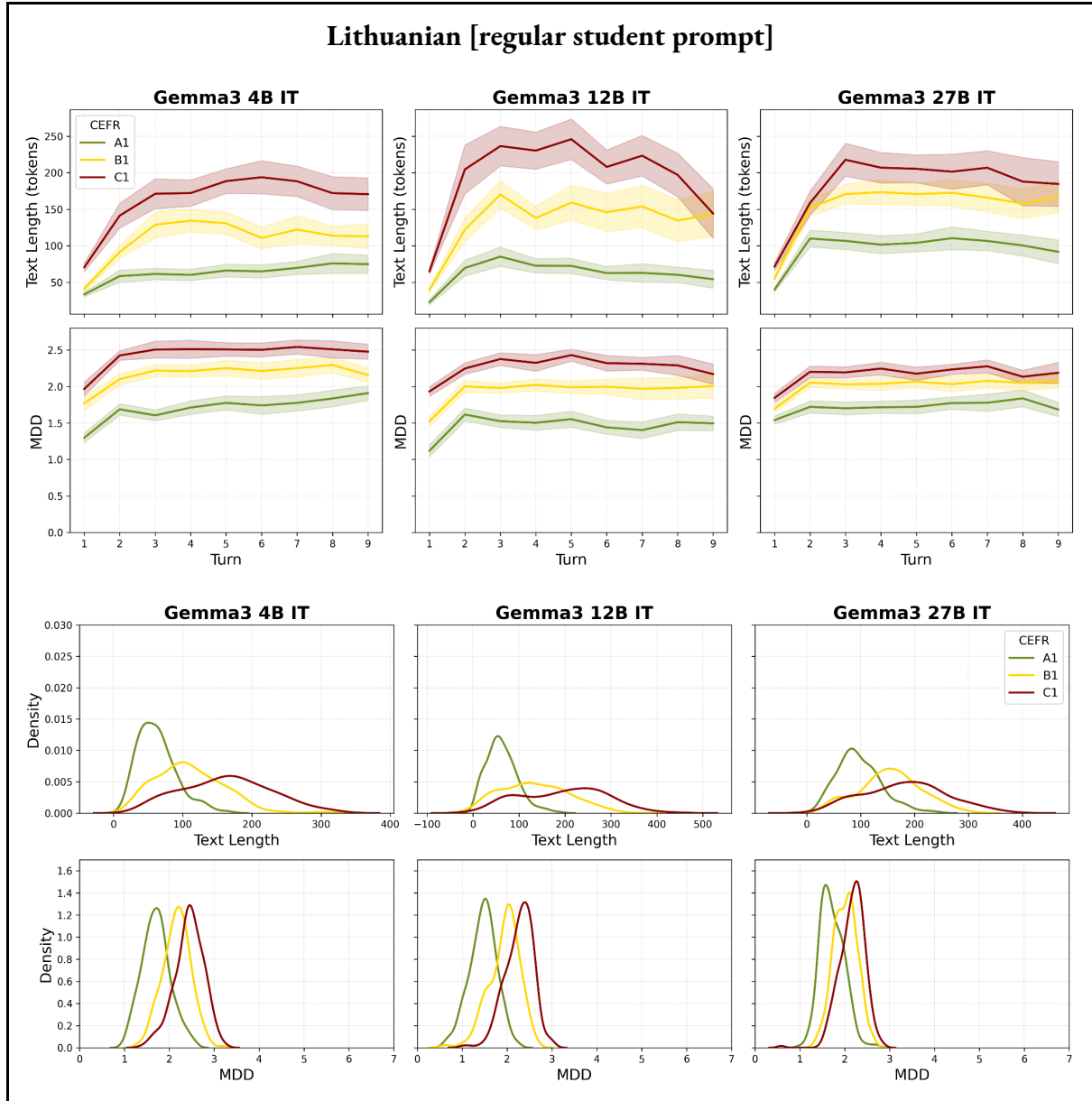


Figure 5: Average text length (tokens) and MDD (mean dependency distance) features extracted from the tutor part of the dialogue when providing the LLM with a system prompt in Lithuanian, using a regular system prompt to simulate the student part of the conversation. The top subplot depicts average values at different time points of the dialogue (across tutor turns). Shaded areas represent 95% CIs. Bottom subplots represent metric densities - the estimated distributions of each metric value across CEFR language proficiency levels. Note: density plots contain different scales.

models appear to generate texts at certain syntactic complexity consistently, as distributions appear relatively narrow. The convergence of values at B1 and C1 levels is depicted in the densities: the distributions for these levels seem to largely overlap, even for the smallest Gemma3 model.

Figure 6-7 addresses RQ 3 by showcasing whether constraining student-LLM behaviour to a specific CEFR level influences the tutor-LLM responses. Generally, the findings across conditions are relatively consistent with previous results: the separation between CEFR levels is clear ($p < 0.001$, see Appendix A.5.3).

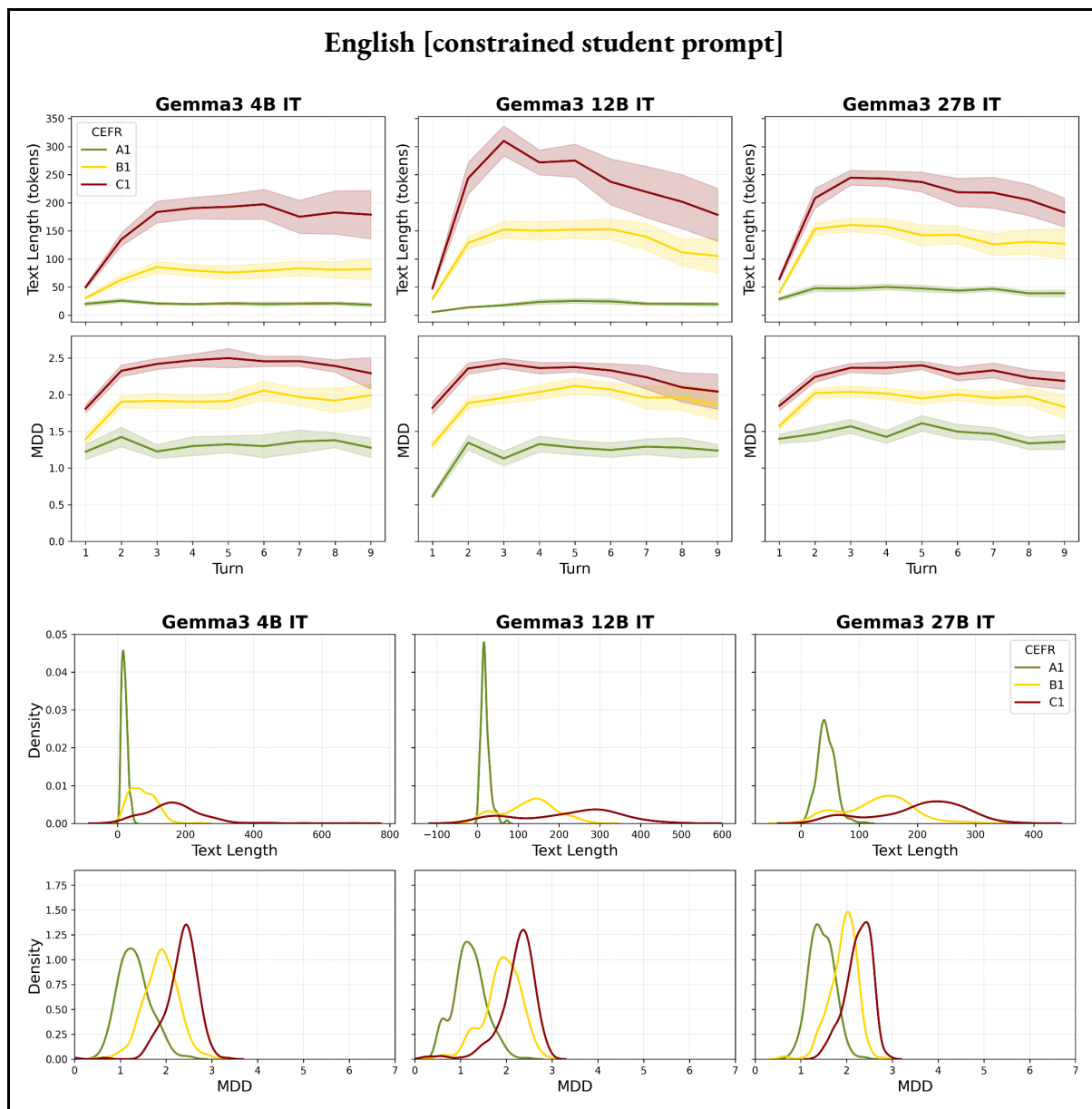


Figure 6: Average text length (tokens) and MDD (mean dependency distance) features extracted from the tutor part of the dialogue when providing the LLM with a system prompt in English using a constraining system prompt to simulate the student part of the conversation. The top subplot depicts average values at different time points of the dialogue (across tutor turns). Shaded areas represent 95% CIs. Bottom subplots represent metric densities - the estimated distributions of each metric value across CEFR language proficiency levels. Note: density plots contain different scales.

The average estimates from linear mixed-effects analysis broadly follow consistent patterns across these two conditions for the corresponding CEFR levels and Gemma3 models (Figure 9). However, the main differences seem to appear within the uncertainty in the estimates/variability of structural features across conversations. Compared to the English system prompting (Figure 6), prompting in Lithuanian (Figure 7) results in more uncertain average estimates throughout the whole conversation, especially at the C1 level.

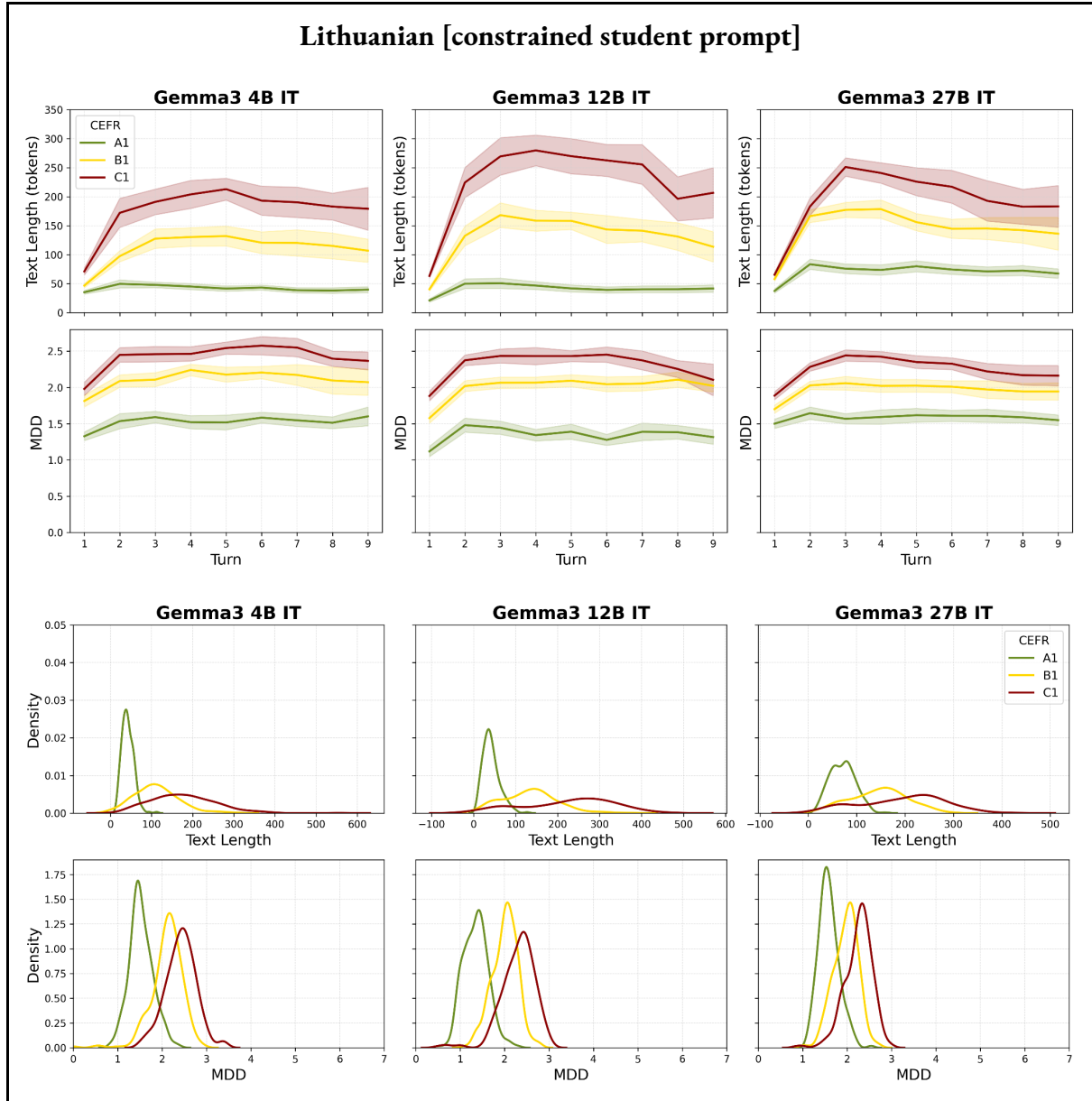


Figure 7: Average text length (tokens) and MDD (mean dependency distance) features extracted from the tutor part of the dialogue when providing the LLM with a system prompt in Lithuanian using a constraining system prompt to simulate the student part of the conversation. The top subplot depicts average values at different time points of the dialogue (across tutor turns). Shaded areas represent 95% CIs. Bottom subplots represent metric densities - the estimated distributions of each metric value across CEFR language proficiency levels. Note: density plots contain different scales.

In both conditions, the convergence of B1 and C1 values towards the end of the conversation is present; however, compared to the previous findings (Figures 4-5), a slight convergence is now visible for the Gemma3 4B model as well. When considering distributions for text length, the English prompting seems to lead to more stable LLM outputs - the distributions are narrower than the ones for Lithuanian. On the contrary, the MDD distributions are slightly narrower for Lithuanian prompting.

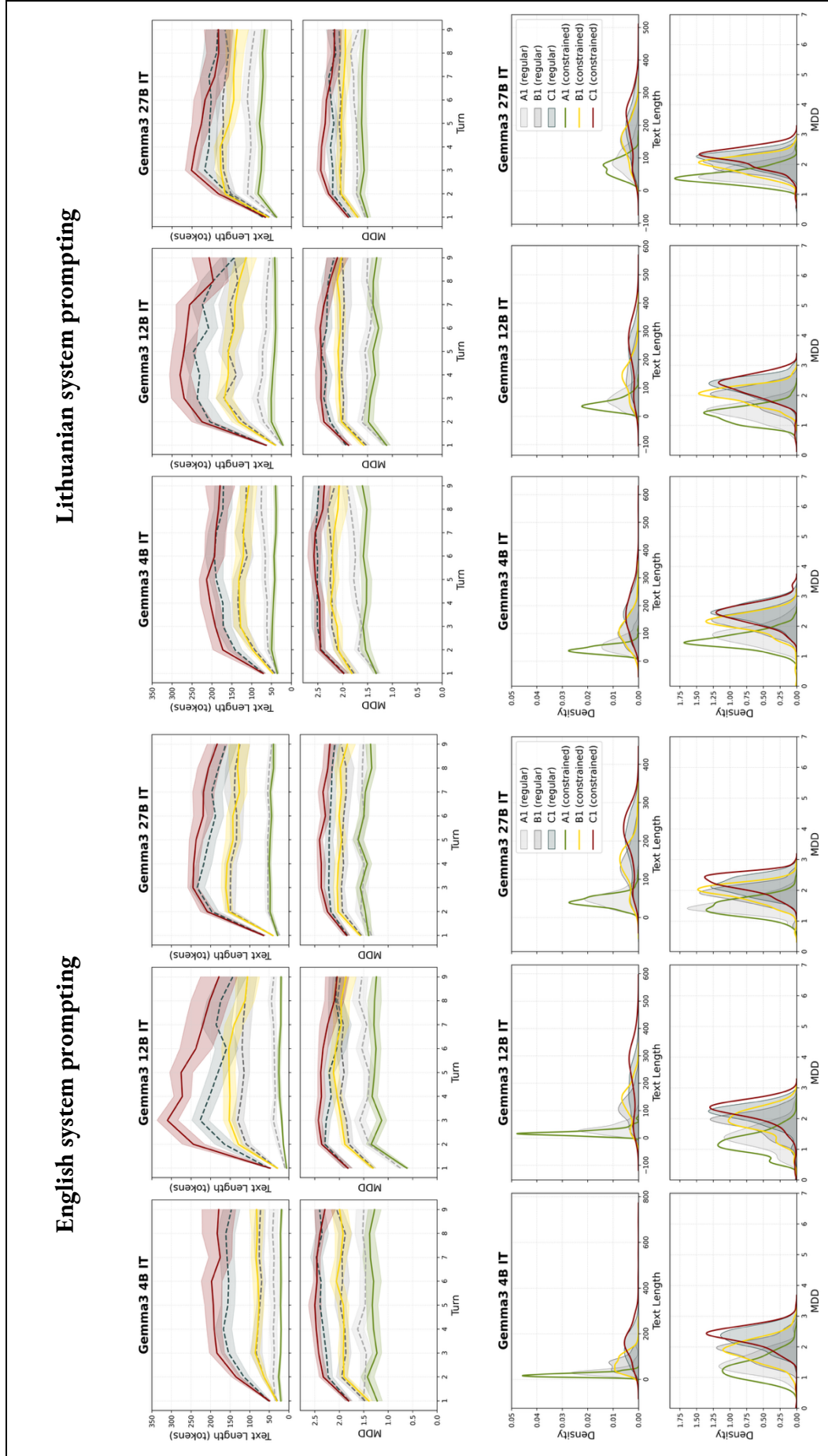


Figure 8: Average text length (tokens) and MDD (mean dependency distance) features extracted from the tutor part of the dialogue. Coloured plots represent metrics extracted from conversations simulated using a constrained student-LLM system prompt. Grey-scaled visualizations - metrics extracted from conversations that were simulated using a regular student-LLM system prompt. Language represents the language for system prompting.

To visually assess the effect of the student-LLM system prompt type on the outcome (text length and MDD estimates within the tutor part of the conversation), language-specific results were combined and displayed in *Figure 8*. Based on the results, it is clear that system prompts that aimed at constraining the student-LLM to a specific language proficiency level increased the distinguishability between CEFR levels even more. This phenomenon is visible in both the English and Lithuanian system-prompting conditions. Nevertheless, turn-level results show a great overlap between the values. Density distributions do not depict major differences in the output either. One of the clearest differences is a slight change in the A1 level distributions: for text length, the distribution becomes narrower, implying slightly more stability in the output; for MDD, the distribution slightly shifts to the left, implying more text of slightly less complexity.

As indicated in previous sections, *Figure 9* showcases the average estimates from linear mixed-effects analysis across the experimental conditions. Based on these findings, the average text length is initially the highest (at A1 - intercept) when providing the LLMs with Lithuanian prompts and supplying the student-LLM with a regular prompt. On average, at the A1 level, Gemma3 4B and 12B models generated text consisting of approximately 63 tokens, whereas the Gemma3 27B model generated 97 tokens. Conversely, the average text length at the A1 level is the lowest overall when prompting the LLMs in English and providing student-LLM with a constraining prompt.

*Appendix A.5.4*⁸ contains the table with aggregated text length estimates across CEFR levels. At the B1 level, findings slightly differ across the models and prompting strategies. Gemma3 12B and 27B models generate the longest text when instructed in Lithuanian using a regular student-LLM prompt. In contrast, the Gemma3 4B model was found to produce the longest responses when the student-LLM is constrained to a B1 level. At the C1 level, constraining student-LLM and prompting in Lithuanian also led the Gemma3 12B model to produce the longest output. Interestingly, the largest Gemma3 model produced the longest text when instructing in English and supplying the student-LLM with a constraining prompt. On the other hand, the shortest outputs in all LLMs at the B1 and C1 levels were produced when prompted in English without supplying CEFR-specific constraints to the student-LLM.

Appendix A.5.5 contains the table with aggregated MDD estimates across CEFR levels. In terms of the MDD estimates, relatively similar results are found. The average estimated MDD values at the A1 level appear to be the highest when LLMs are instructed with Lithuanian system prompts and provided with a regular system prompt for the student-LLM. Interestingly, the Gemma3 4B and 27B models were estimated to produce text of relatively similar syntactic complexity - 1.70 and 1.72, respectively. The lowest MDD value at the A1 level was estimated when models were prompted in English, using a constraining prompt for the student LLM.

⁸ Visualizations of aggregated average estimates are included in Appendix, sections A.5.1 – A.5.3.

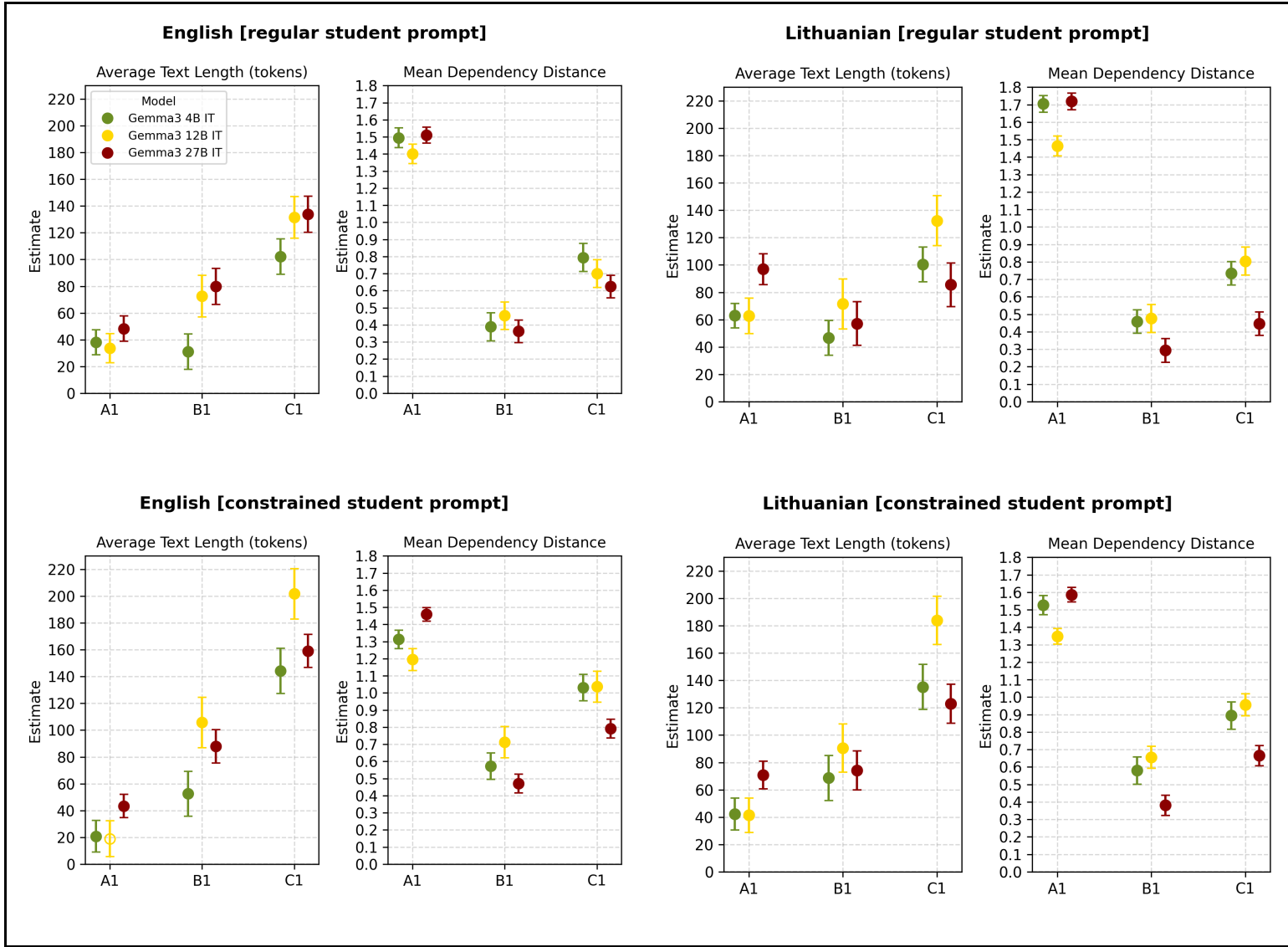


Figure 9: Comparison of average estimates from linear mixed-effects analysis across the Gemma3 4B, 12B, and 27B models. The x-axis refers to CEFR levels: A1 (intercept/baseline), B1, and C1. The y-axis represents the average estimates at respective CEFR levels. The y-axis refers to the estimated values for each linguistic metric (average text length and mean dependency distance, namely, MDD). The filled markers represent statistically significant estimates, otherwise insignificant. Estimates for B1 and C1 represent an incremental increase from A1 (intercept). Note: all tables with respective estimates can be found in Appendix A.5.

At higher CEFR levels, results generally match the ones described for text length: the LLM outputs lead to higher MDDs when prompted in Lithuanian, and the lowest when prompted in English, especially when a regular student-LLM prompt is used.

4.2.2 Text Complexity Analysis: Tutor Texts vs. Reference Data

Syntactic and morphological features were extracted from the tutor part of the dialogue across the conditions. Findings were compared against the reference dataset and visualized in Figures 10-11. Overall, across all conditions, at each CEFR level, Gemma3 models were found to generate slightly fewer finite verbs compared to the finite verb frequencies in the reference data (Figure 10).

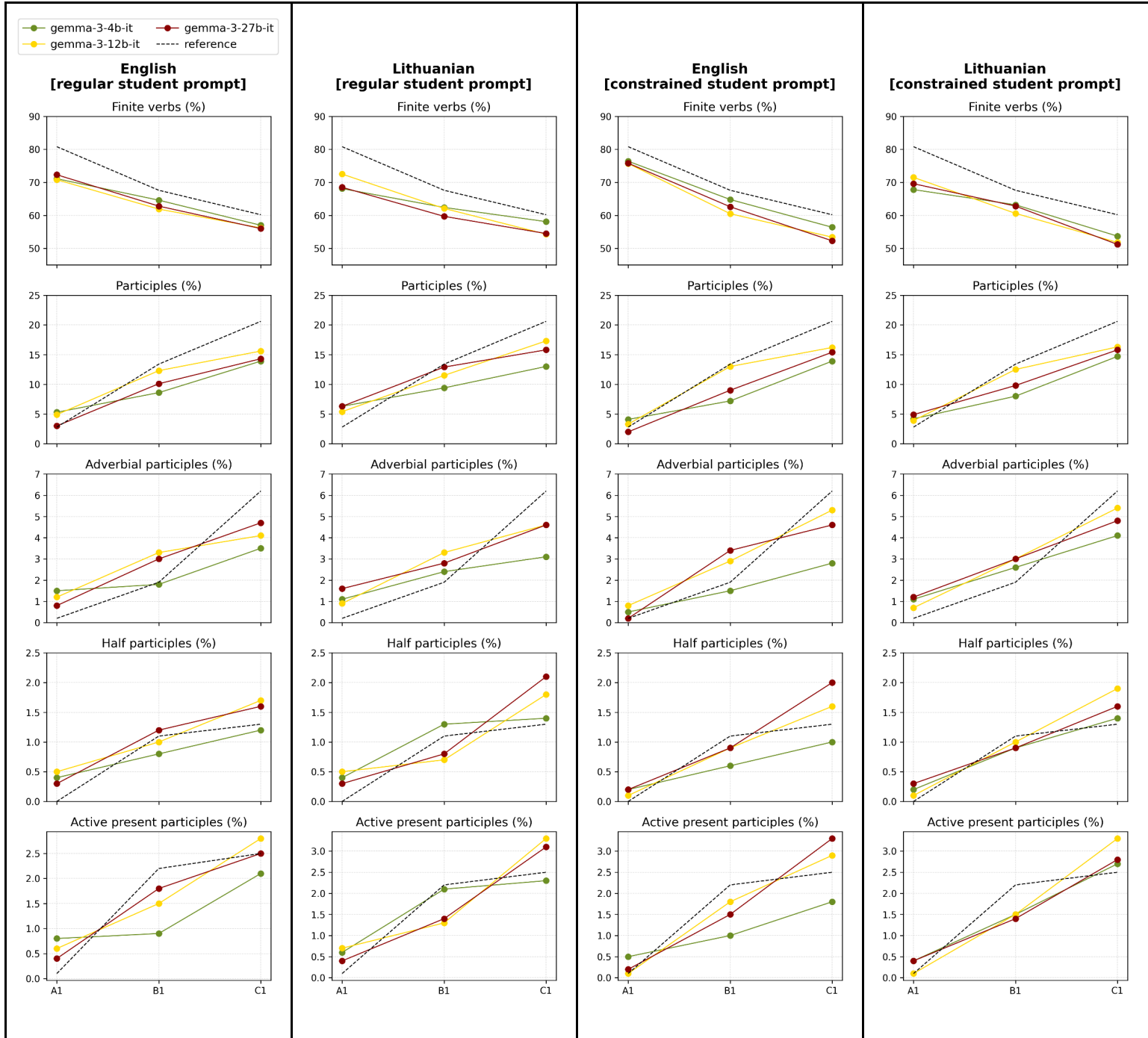


Figure 10: Verb-related morphological features in tutor-LLM versus reference texts. The dashed line represents the frequencies in the reference data. The y-axis represents relative frequencies (%) within the specific verb category. The x-axis refers to the CEFR levels.

As expected, the frequency of finite verbs seems to decrease at higher proficiency levels. Compared to the reference text, simulated texts contain a higher frequency of participles at the A1 level, whereas at higher levels, frequencies appear to be lower. This trend is similar across all models; however, the Gemma3 12B model seems to resemble the reference data the closest. Results regarding the adverbial and half-participles suggest similar trends: in most cases, LLMs frequencies are higher at the A1 level (Figure 10).

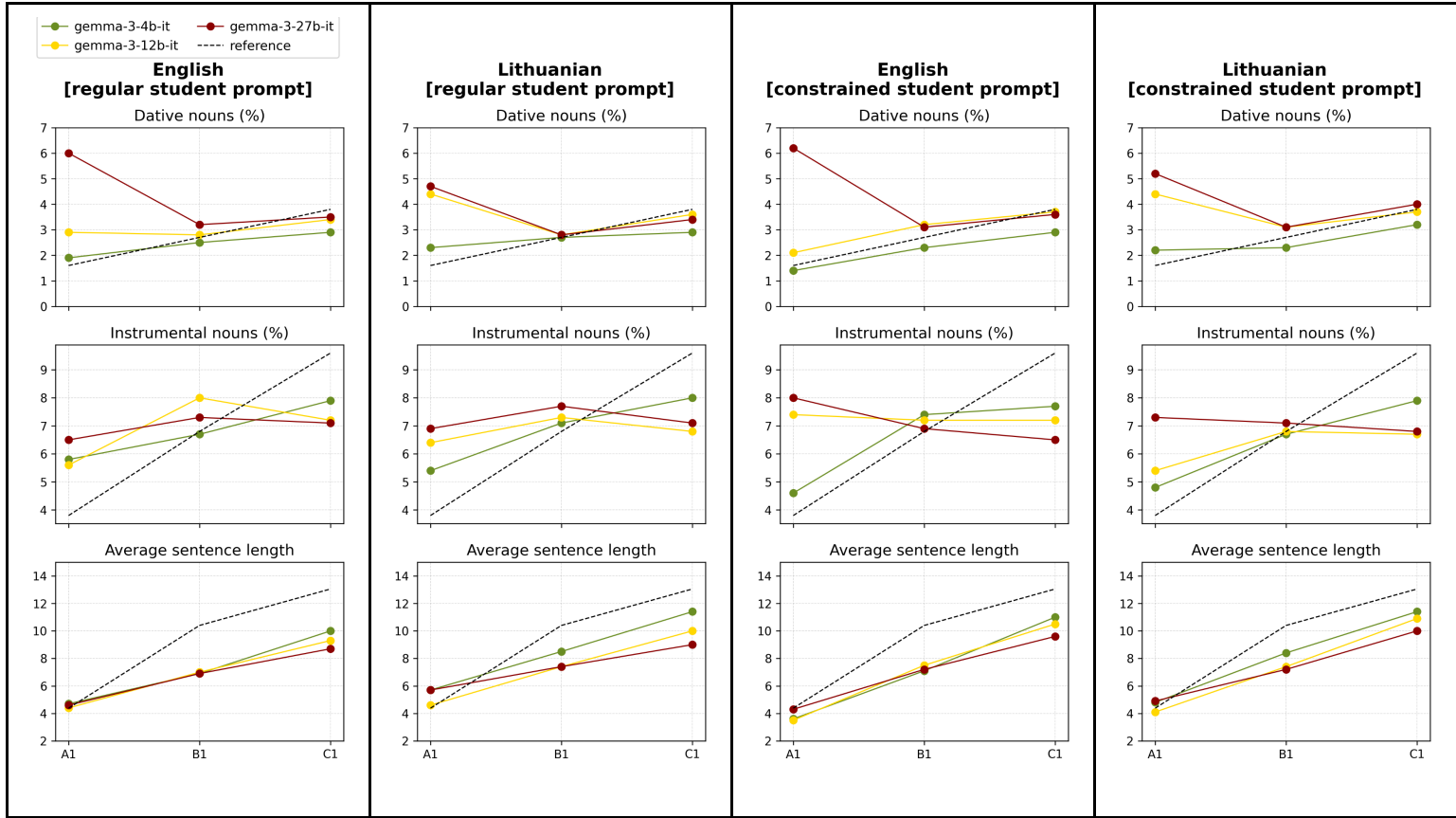


Figure 11: Noun-related morphological features and sentence length in tutor-LLM versus reference texts. The dashed line shows reference frequencies. The y-axis represents relative frequencies (%) within the specific noun category and mean sentence length. The x-axis refers to the CEFR levels.

For adverbial participles, Gemma3 models tend to have higher frequencies at B1 and lower frequencies at C1 levels. While a similar trend is captured in the plots with frequencies of half-participles, generally, all models tend to produce them more frequently than found in reference texts. The latter findings are also depicted in the plotted frequencies of active present participles as well.

In Figure 11, the frequencies of dative and instrumental nouns are a lot higher at A1 level simulated texts, especially for Gemma3 12B model when prompting is performed in Lithuanian, and 27B model across all conditions. For both types of nouns frequencies are relatively similar to the ones in the reference data at B1 level across all models. At the C1 level, the frequency of dative nouns in simulated texts resembles the reference data closely; however, instrumental nouns are less frequent. Based on average sentence length, the general pattern across the conditions appears to be similar: at the A1 level, all models tend to match the average sentence length closely to that of the reference data, however, the reference sentences at the higher CEFR levels are, on average, longer than the ones in the simulated texts. When comparing findings between conditions, it appears that the sentence length in the Gemma3 4B model responses matches the reference slightly more when prompted in Lithuanian as opposed to prompting in English. Finally, MDD metrics were extracted from the

reference texts. On average, at the A1 level, the estimated MDD was approximately 1.98, whereas at the B1 and C1 levels, 2.53 and 2.63, respectively.

4.2.3 Distance-based Analysis of Alignment

Figure 12 shows the absolute average MDD distance between the simulated tutor-LLM and student-LLM responses over time. Overall, it appears that there is always a difference between the tutor-LLM and the student-LLM texts, with the distance on average being largest at the A1 level and slightly smaller at the B1 and C1 levels. The turn-level dynamics showcase an early alignment in the MDD distance between the tutor and a student. For instance, when prompted in Lithuanian without student-LLM constraints, the Gemma3 12B model showcases a large MDD difference in the generated texts at A1 and B1 levels during the 1st conversational exchange. The latter is also visible in both student-LLM constraining conditions for Gemma3 4B and 12B models. From the second turn onwards, the models appear to generally reach a somewhat equilibrium - the distance drops and becomes more stable. In other cases, the conversation starts with a rather small difference in MDD, and as the dialogue progresses, the distance increases and stabilizes.

Gemma3 4B model appears to be the least stable with respect to the stability in the interaction between the tutor and the student. When prompted in Lithuanian with no student-LLM constraints, there appears to be a sharp increase in the MDD difference at the C1 level, in the middle of the conversation, suggesting that one of the LLMs kept diverging from the instructions provided. While 95% CIs do not overlap at the A1 level in some cases, most often there is a great overlap across all CEFR levels. As such, findings suggest that the underlying MDD distributions of the absolute average distances across the conversations are relatively similar.

Figure 13 showcases the absolute average distance in text length of tutor-student conversations. As opposed to the absolute average MDD distance findings (*Figure 12*), the largest distances/differences in text length are observed at the C1 level, whereas the smallest - at A1. As such, text length between the tutor and student-LLM is similar at the A1 level and quite different at C1. In all cases, the 95% CIs are quite narrow for A1 level distance distributions, suggesting relatively small variability in differences of text length across the conversations. However, at higher language proficiency levels the variability seems to increase.

Although it appears that the difference between tutor- and student-LLM outputs is always present, some instances might indicate a tendency for the models to generate text of similar length. As an example, the distance in text length for Gemma3 12B model at B1 and C1 levels appears to decrease towards the end of the conversations, indicating that the tutor- and student-LLM outputs were becoming more alike. Such a pattern is also present in Gemma3 27B model outputs at the C1 level, especially when prompted in English.

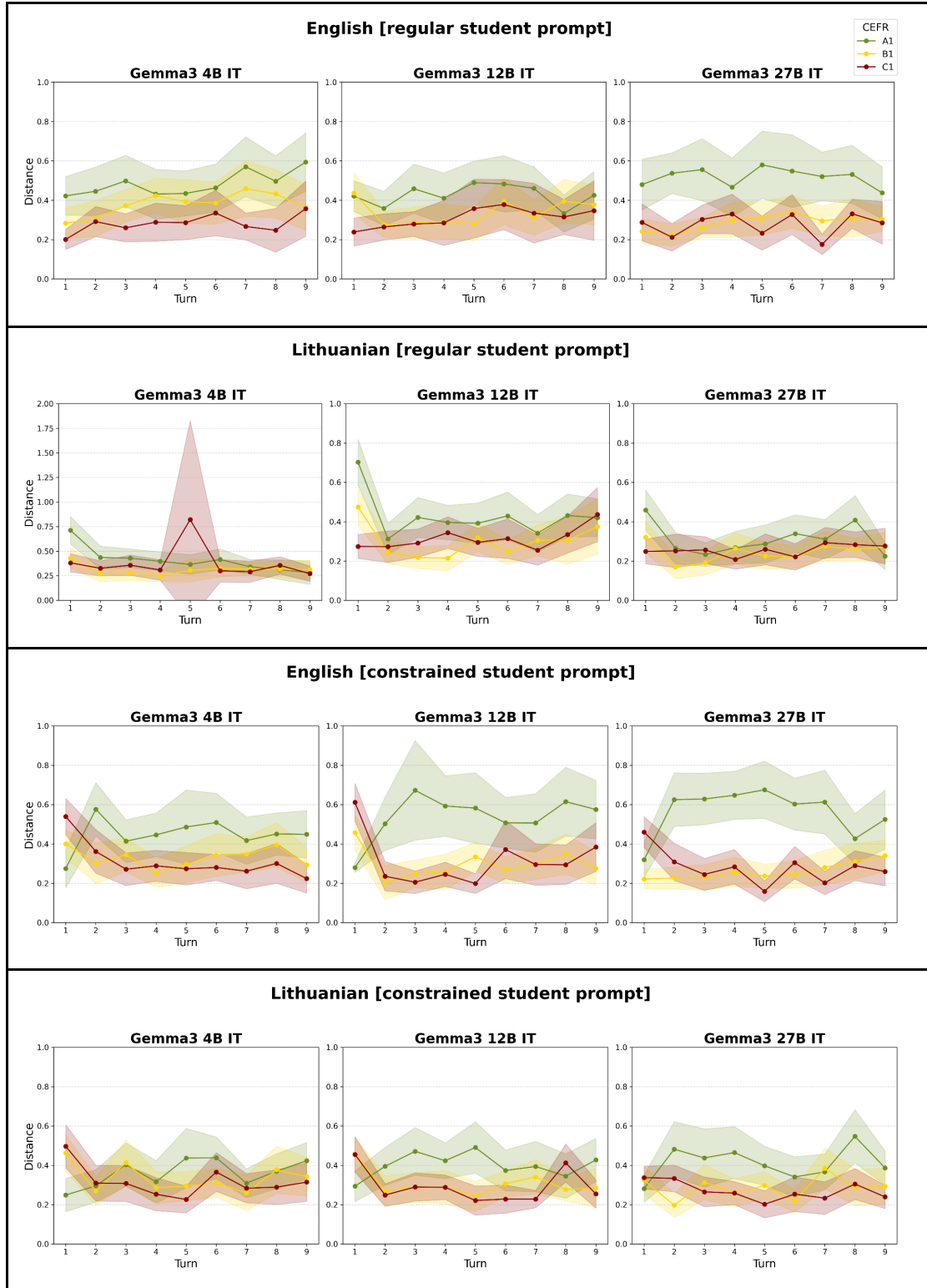


Figure 12: The absolute average MDD distance between the tutor- and student-LLM across conversation turns, for each experimental condition. Shaded areas represent 95% CIs. Note: different scale for Gemma3 4B when prompted in Lithuanian and supplied student-LLM with regular system prompt.

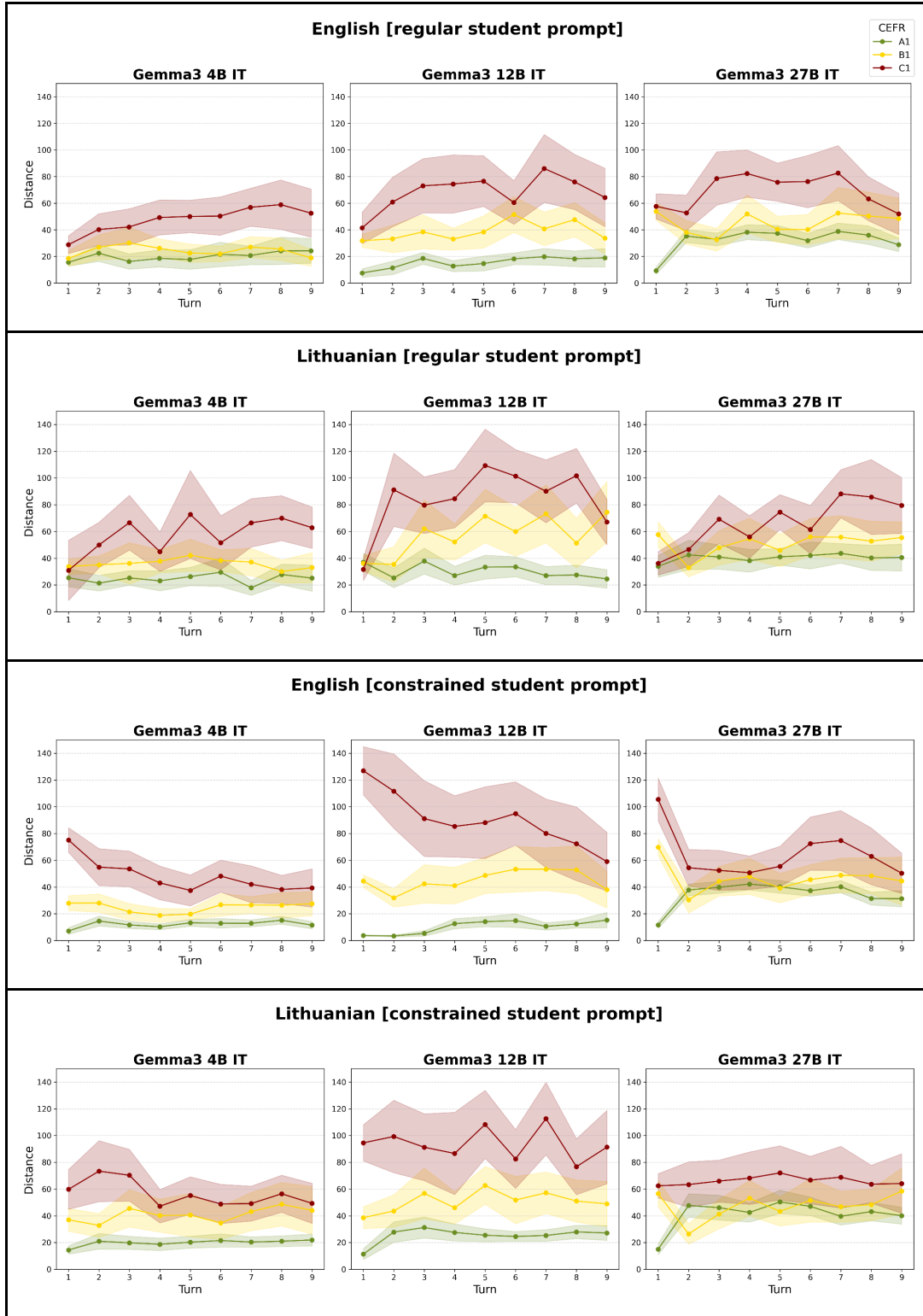


Figure 13: The absolute average distance in text length between the tutor- and student-LLM across conversation turns, for each experimental condition. Shaded areas represent 95% CIs.

4.2.4 Mitigating the Alignment Drift

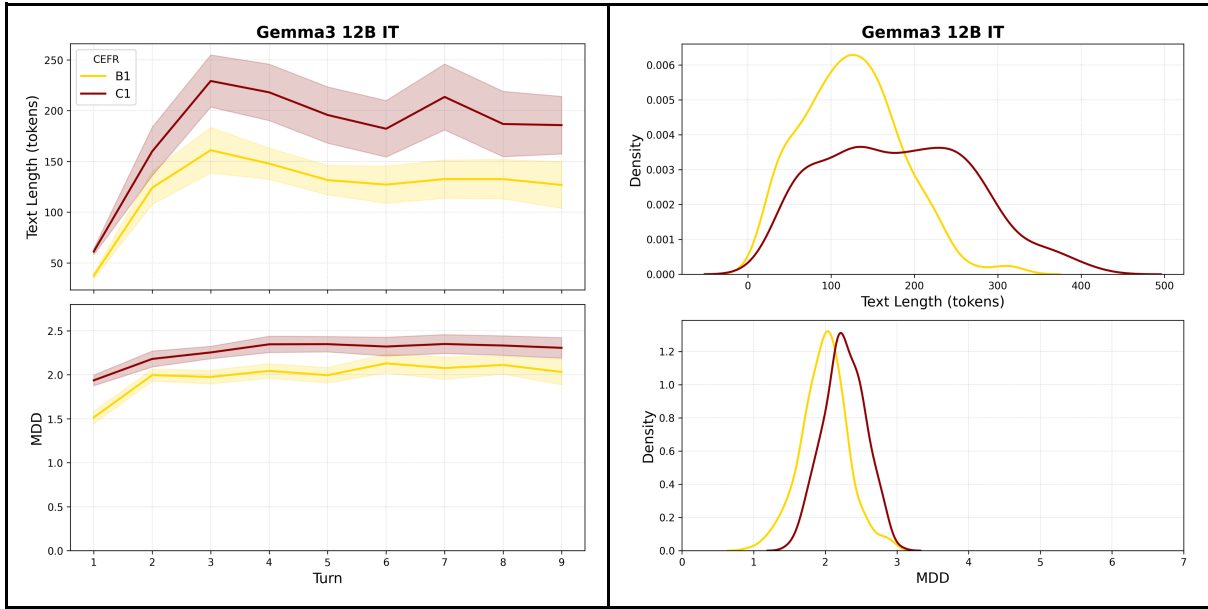


Figure 14: Average text length (tokens) and MDD features extracted from the tutor part of the dialogue when providing the LLM with a system prompt in Lithuanian, using a regular system prompt for simulating the student part of the conversation. Left-side subplots depict average values at different time points of the dialogue (across tutor turns). Shaded areas represent 95% CIs. Right-side subplots represent metric densities - the estimated distributions of each metric value across CEFR language proficiency levels. Note: density plots contain different scales.

Figure 14 refers to RQ 4 by showcasing how the new experimental design, introduced in section 3.6, improves the alignment drift by instructing the tutor-LLM to self-refine the generated text once it starts diverging. The choice of language for system prompting, the type of system prompt used to instruct the student-LLM and the language model itself to generate the conversations was based on the previous findings, namely, the quality of the generated text (section 4.1.1), the lexical validity of words (4.1.2) and the overall results of the alignment drift (section 4.2.1). Moreover, the computational resources needed to run additional simulations were also considered.

Considering findings in Figure 2 (section 4.1.1), the number of English sentences in the tutor-LLM output was substantially reduced when the models were supplied with system prompts in Lithuanian. This is important for experimental design, as cleaner responses might provide a more basis for computing structural complexity metrics. The variability in error-rates was also found to be slightly lower when the student-LLM was not constrained to a specific CEFR level (Figure 3, section 4.1.2). Moreover, when considering section 4.2.1 findings in relation to prompting in Lithuanian the findings of alignment drift are slightly more evident when the student-LLM is not constrained to a CEFR level (Figure 5), especially for the Gemma3 12B model. A stronger initial alignment drift was expected to facilitate an assessment of whether the new framework is effective in improving it. Lastly, tutor-LLM responses at turns 3 and 4 were selected as references for adapting subsequent turns. Therefore, the experimental design was run using system-prompting in

Lithuanian, without constraining student-LLM to a CEFR level, using a Gemma3 12B language model, and selecting turns 3 and 4 as reference points.

Compared with the previous findings, namely, *Figure 5*, it is clear that the average text length at the C1 level does not converge to the estimated values at level B1. Moreover, the 95% CIs appear to overlap slightly only at the start of the conversation. The density distributions for text length show slight changes as well - the distributions are narrower, especially for text length at the B1 level, indicating less diverging text length in some of the LLM outputs. When comparing the average MDD values, it seems that there appears to be less value convergence across the CEFR values as well. While the 95% CIs are slightly more overlapping, the average values at the C1 level do not show a tendency to converge. The density distributions for MDD are also slightly narrower, especially at the C1 level: all texts appear to have the MDD estimate of at least 1.

5. Discussion

The findings of the present study were intended to reveal whether the phenomenon of alignment drift is present in tutor-LLM responses when the language model is instructed to act as a dialogue partner to a student who is learning Lithuanian as a second language (L2). The findings of the study might be useful in supporting the development of tools for learning Lithuanian as L2. The study followed an experimental design employed by Almasi & Kristensen-McLachlan (2025) and was further expanded to address some of the limitations that were indicated as potential reasons for alignment drift to occur. The sections below contain the discussion of the findings.

5.1 Alignment Drift

5.1.1 Alignment Drift in Lithuanian Conversations

RQ 1 refers to the general investigation of alignment drift in simulated tutor-student conversations for Lithuanian language learning. Based on previous research on the Lithuanian language (Kapočiūtė-Dzikiene et al., 2025), the EuroEval benchmarking results, and the language models used by Almasi & Kristensen-McLachlan (2025), the Gemma3 4B, 12B, and 27B models were employed to generate tutor-student conversations in Lithuanian.

As in the study by Almasi & Kristensen-McLachlan (2025), it was found that all Gemma3 language models were able to generate tutor responses which differed depending on specific CEFR language proficiency level, as initially intended. The extracted metrics of average text length and mean dependency distance were found to increase with higher CEFR levels, implying more structurally complex text. When comparing findings derived from the Gemma3 12B model when prompted in Spanish (Almasi & Kristensen-McLachlan, 2025), system-prompting in Lithuanian led to a substantially clearer separation between CEFR levels across the turns in the conversation, especially

for MDD metrics. However, as originally found by the authors, alignment drift appeared in Lithuanian conversations as well.

While the alignment drift was not as severe, the Gemma3 12B and 27B models were found to drift towards the end of the dialogue (*Figure 4*) in terms of both average text length and syntactic complexity. The tendency to drift appears to begin quite early in the conversation, and only at the B1 and C1 levels. From approximately turn 6 onwards, the average text length and syntactic complexity of the LLM-generated tutor output appeared to become quite similar, suggesting that system prompting alone is not powerful enough for instructing larger Gemma3 models to act as an L2 tutor at B1 and C1 language levels. Gemma3 4B model, on the other hand, did not appear to show value convergence between the B1 and C1 levels. There might be a tendency for the Gemma3 4B model to produce longer B1 level texts from turn 8 onwards; however, as simulations are limited to nine turns only, it is unknown whether the increase in text length would continue and eventually converge to C1 level text lengths.

Overall, it is evident that the alignment drift found by Almasi & Kristensen-McLachlan (2025) is not language-specific, as it is present in Lithuanian conversations. Moreover, a recent study by Jin et al. (2026) showcased the alignment drift being present in Japanese conversations as well. As such, these findings confirm that the system prompting by itself might not be able to constrain the tutor-LLM behavior to a specific CEFR level for the whole course of interaction with a student.

5.1.2 System Prompting in Lithuanian

RQ 2 refers to the investigation of whether the tutor-LLM responses can be improved when system-prompting is done in Lithuanian as opposed to English. To achieve this, system prompts in English were translated into Lithuanian by a native speaker. The CEFR-specific part of the system prompt was replaced using an official translation from one of the Lithuanian language schools.

When prompted in English, the Gemma3 4B model did not show a clear drift in CEFR level alignment (*Figure 4*). This finding generally holds for the condition where prompting was performed in Lithuanian as well (*Figure 5*). However, the A1 level responses appear to show a tendency to drift toward higher structural complexity, as the average metrics increase over time and move closer to the B1 level estimates. Structural complexity at A1 and B1 levels appear to be higher in general, suggesting that tutor-LLM responses at lower levels when prompted in Lithuanian become more complex, while the complexity of C1 level responses remains relatively the same.

The average metrics did not show much of the improvement in Gemma3 12B model output. Although prompting in Lithuanian seems to slightly delay the observed drift in text length at the C1 level, the phenomenon still reappears. The extracted MDD values from the tutor-LLM responses indicate slightly less convergence in B1 and C1 level values as well, however, due to the increased

variability of MDD values towards the end of the conversation, the estimates at B1 and C1 levels end up converging.

The findings for the Gemma3 27B model indicate a clearer improvement in alignment when prompted in Lithuanian as opposed to English. The average text length and MDD estimates do not suggest that the tutor-LLM responses become structurally simpler toward the end of the conversation. Instead, the relatively stable values over time indicate that the Gemma3 27B model generally maintains a similar level of structural complexity in its responses across the turns. As for the Gemma3 4B model, the Gemma3 27B model responses are found to generally become more complex when system-prompting is performed in Lithuanian rather than English.

Overall, supplying language models with system prompts translated to the Lithuanian language appears to improve the responses of the Gemma3 27B model slightly. However, alignment drift becomes visible in the Gemma3 4B model responses, where it was not evident under English system-prompting (*Figure 4*). For the Gemma3 12B model, Lithuanian system-prompting does not appear to substantially change the response patterns. Generally, when models are instructed in Lithuanian, A1 and B1 level responses become more complex overall. As such, complexity in B1 and C1 level responses becomes more alike.

5.1.3 Constraining the Student to the Specific CEFR Level

RQ 3 refers to the investigation of whether the tutor-LLM adaptation to a specific CEFR level can be improved if the student-LLM is constrained to the respective language proficiency level. It was hypothesized that due to student-LLM being unconstrained, the language model might end up generating too simple or too complex text that could potentially influence the tutor-LLM responses, resulting in alignment drift. Since the language used for system prompting might influence the LLM responses already, simulations of conversation between a tutor- and a student-LLM were performed in both English and Lithuanian languages. To answer this research question, a new set of system prompts was developed for constraining the student-LLM to a specific language proficiency level.

Based on the findings presented in *Figures 6, 7, and 8*, constraining the student-LLM did not improve the tutor-LLM drifting towards the end of the conversation. However, it appears that the overall differentiation between the CEFR levels improved - the distances between the average text length and MDD values at A1, B1, and C1 levels are larger. Across all models and in both prompting languages, variability remained relatively low during the first half of the conversations, indicating that the tutor-LLM consistently produced responses of a similar text length and syntactic complexity. In fact, the uncertainty in responses appears to be slightly lower compared to the previous findings, especially when considering all model responses at the A1 level.

Regarding the drift in CEFR level alignment, it seems that the model constraints not only fail to improve the output, but it also resulted in a more severe tutor-LLM drift overall. The uncertainty in

estimates towards the end of the dialogues increased, in some cases resulting in the overlapping value distributions at the B1 and C1 levels. Without student-LLM constraints, it appears that the tutor-LLM's responses would drift mainly at the C1 level, subsequently converging with the B1 level values. However, with student-LLM constraints, the drifting is present in both B1 and C1 levels, meaning that once the student is constrained to a specific proficiency level, the tutor tends to produce simpler B1 and C1 level texts. The latter is true for Gemma3 12B and 27B models, although the Gemma3 27B model appears to be slightly less prone to drift at the B1 level. The findings for the Gemma3 4B model, however, showed relatively little overlap between the CEFR levels. Nevertheless, compared with the previous findings, syntactic complexity at the C1 level appeared to fluctuate more and become increasingly similar to that observed at the B1 level.

Overall, constraining the student-LLM did not improve the phenomenon of alignment drift. Language for system prompting did not appear to influence the outputs either. Due to student-LLM constraints, the alignment drift became more prevalent, with average estimates decreasing not only at the advanced, but also at the intermediate level. Moreover, compared with previous findings, the phenomenon became more prevalent in Gemma3 4B model responses as well.

5.2 Alignment Drift Versus Mutual Adaptation

An exploration was established to find out whether the observed phenomenon is in fact a drift in alignment, or it is an effect of two language models adapting to one another, namely, the result of mutual adaptation (section 3.5.6). *Figures 12 and 13* showcase the absolute average distances in the computed MDD and text length metrics, respectively, between the tutor- and student-LLM. Generally, the average distances do not imply that the LLMs managed to reach a clear mutual adaptation over the course of interaction, as the average difference between the models across the turns is always above zero. However, there might appear patterns in the tutor- and student-LLM interactions that might indicate an increase in output similarity, potentially implying some degree of mutual adaptation over tutor-student exchanges.

A pattern of mutual adaptation might be captured by a notable decrease in the distance of text length in Gemma3 12B and 27B model simulated C1 level responses, for cases where instructions were provided in English. Such a decrease indicates a potential for tutor-student interactions to gradually become similar. *Figures 4 and 6* indicate that tutors' responses generated by respective language models at the C1 level on average became shorter towards the end of the conversations. Taken together, these findings may imply that once the instructions to the models are provided in English, the models tend to reciprocally adapt to one another by gradually matching one another's responses in terms of the text length, potentially indicating some degree of mutual adaptability.

In contrast, notable oscillations in the distances in text length between the tutor- and student-LLM are depicted in the model responses produced under Lithuanian prompting. *Figure 13* depicts these

oscillations most prominently for C1 and sometimes B1 level metric distances. Moreover, the oscillations, i.e., the variability in distances across turns, are evident across all Gemma3 models, particularly when the student-LLM is not constrained to a specific CEFR level. While *Figures 5 and 7* still demonstrate a general decrease in tutor response length over time, the oscillating distance patterns might imply the reduced overall model stability in conversational dynamics under Lithuanian prompting. Importantly, when CEFR-specific constraints are supplied within the student-LLM system prompt, fluctuations in distances are less pronounced (*Figure 13*). This observation reinforces the possibility that the student-LLM model stability is more dependent on explicit instructions provided in Lithuanian settings.

Finally, the variability in average distances of A1 level text length appears to be relatively stable across the turns. Moreover, there seems to be no notable difference in distance distributions, neither across the models nor the experimental conditions. As such, it might be that beginner-level system prompts were stronger in constraining tutor- and student-LLM's behaviour by limiting the extent to which the LLMs could diverge.

The distributions of MDD distance (*Figure 12*) appear much more stable across the models and experimental conditions. The stability might imply that tutor- and student-LLMs maintained a consistent degree of syntactic complexity throughout the interactions. Contrary to the findings reported by Kandra et al. (2025), LLMs did not show neither mutual adaptiveness nor diverged from one another in terms of syntactic complexity across the turns. As such, compared with the text length, syntactic complexity might in general be a more stable property within tutor-student interactions.

Overall, the absolute average distances computed between the tutor- and student-LLM responses did not provide clear evidence that the observed alignment drift in the tutor-LLM responses resulted from mutual adaptation between the tutor and student language models. While some degree of reciprocal adaptation might be inferred, drawing concrete conclusions based on absolute distances should be avoided, as absolute differences on their own do not depict the underlying interactional dynamics between the tutor- and the student-LLM. Nevertheless, it is clear that the LLMs did not converge to outputs of identical structural complexity.

5.3 Effectiveness of Alignment Drift Mitigation

RQ 4 refers to the exploratory investigation of how/to what extent alignment drift in tutors' responses can be improved if a new experimental design, informed by previous findings, is introduced to the study. Based on the RQ 1-3 relevant findings, it was found that switching to the Lithuanian system-prompting improved the alignment for Gemma3 27B model slightly, whereas drift in alignment appeared to be prevalent in some Gemma3 4B and Gemma3 12B tutor-LLM responses. Moreover, constraining the student-LLM to a specific proficiency level slightly amplified

the presence of the drift. Nevertheless, although four experimental conditions (section 3.3.3.4) were tested and some outcome differences were identified, the tutor-LLM responses remained relatively similar across the different settings, without substantial variation.

Surprisingly, the smallest model, namely, Gemma3 4B, was the least susceptible to alignment drift, supporting recent observations that larger LLMs are not always better at the given task (Guo et al., 2026; Li et al., 2025). Larger LLMs, namely, Gemma3 12B and 27B models, were found to struggle to adhere to instructions throughout the whole dialogue, especially at levels B1 and C1. One potential reason for this could be rather structurally simple Lithuanian data that the Gemma3 LLMs were trained on. As such, larger LLMs, such as Gemma3 12B and 27B models, might initially respond stronger to the instructions provided in the system prompt and therefore generate more complex text at the beginning of the dialogue. However, as the interaction between the tutor and the student progresses, the influence of system prompting might become weaker, causing LLMs to regress toward the text generation patterns similar to those that they were trained on. As a result, the complexity estimates at higher levels end up converging to the ones produced by the Gemma3 4B model or slightly lower (*Figure 4-8*). Based on findings in section 5.2, there is no supporting evidence that the observed alignment drift results from mutual adaptability in tutor-student response complexity. Rather, alternative explanations might include misalignment in tutor- or student-LLM responses in prior turns, or as previously mentioned, the diminished influence of system prompts (see section 3.6). For these reasons, an adapted self-refinement framework was tested (Madaan et al., 2023).

Figure 14 depicts the findings from the newly proposed experimental design. Based on the structural complexity metrics, Gemma3 12B model's output appears to be greatly improved: there is no visible drift towards the end of the conversations; moreover, the distributions at B1 and C1 overlap only slightly. However, during text generation, the model quite often failed to produce the response for which the extracted structural complexity metrics would fall within the conversation-specific metric thresholds. As such, most often than not, the LLM would exhaust the allowed regeneration/self-refinement attempts, and therefore the last, although invalid attempt was saved. Moreover, in most cases, the tutor-LLM would produce an apologetic response after the instruction of refinement was provided, such as:

“Atsiprašau už tai. Mano klaidos mechanizmas kartais leidžia nukrypstant nuo numatomo lygmens.”

Such an LLM's apology as a response to a corrective prompt appeared quite often in the model's output, and therefore, was a part of the evaluation. As such, although self-refinement did improve the model's ability to generate relatively stable responses, the apologetic and self-referential language became a limitation of such an approach.

5.4 LLM-Generated Lithuanian Text

5.4.1 Quality Assessment

While previous sections concerned the investigation of how well the language model is able to adapt and align with CEFR level-specific instructions, the quality of the generated text is important to assess as well. Sections 4.1.1 and 4.1.2 presented how the experimental conditions influence the output quality of the different Gemma3 language models.

Gemma3 4B model, although it did not show severe alignment drift patterns, was found to produce repetitive token sequences more frequently than larger Gemma3 models (*Figure 2*). Additionally, in most of the cases, the model appeared not to be able to recover from the repetitive generation loop state. The language model was also found to produce the highest rate of spelling errors across conversations, particularly within the beginner level output (*Figure 3*). Compared to larger Gemma3 models, the high frequency of lexically invalid words might be an impact of substantially less data that the model was trained on, and therefore, it is quite possible that the Lithuanian language is proportionally less represented, leading to a higher rate of invalid Lithuanian words (Kamath et al., 2025).

Taking all of the experimental conditions together, Gemma3 12B and 27B outputs were found to be somewhat alike, suggesting that larger parameter size and broader training data coverage do not necessarily improve the quality of the generated text. Relatively poor text generation aligns with recent findings that multilingual LLMs, such as the Gemma3 27B model, simply struggle to produce accurate texts in languages that are typologically distant from English (Arčon et al., 2026). Moreover, LLMs were shown to struggle to internalize morphological constraints, particularly for morphologically rich languages, resulting in lexical validity errors within the generated text (Ismayilzada et al., 2025).

Overall, the Gemma3 4B, 12B, and 27B instruction-tuned models are relatively poor in generating clean, lexically valid Lithuanian text. As such, using these models as tutors for L2 learning can be particularly problematic, as language learners might be at risk of being exposed to a number of errors within the text.

5.4.1 Proficiency Assessment

Figures 10 and 11 showcase verb- and noun-related morphological feature frequencies in tutor-LLM versus reference texts. In general, it appears that all language models follow a pattern of increasing complexity as CEFR levels get higher. However, compared to the pedagogic (i.e., reference) texts, complexity does not always match as intended. Verb-related morphological feature frequencies (*Figure 10*) at the A1 level are always higher than those in reference texts, implying that all language models produce A1 level text that is more complex than the one used to teach beginner-level

language learners. Findings are more mixed at higher levels - in some cases, the LLM output is too simple, in others, too complex.

Somewhat similar patterns hold for noun-related morphological feature frequencies (*Figure 11*) as well. However, the difference in A1 level noun frequencies within the LLM versus reference data is much more pronounced. In fact, the frequencies across CEFR levels for each Gemma3 model are approximately alike, suggesting that CEFR-based prompting might not have an influence on noun-related morphological feature frequencies, and the LLM's inherent behaviour appears to show more dominance.

The average sentence length differences imply that, in general, LLM-generated sentences at the A1 level match the A1 level sentence lengths in the pedagogic dataset. However, simulated sentences at higher levels were found to be consistently shorter. When considering syntactic complexity, it is clear that, on average, syntactic complexity is higher in the reference data. *Figures 4-7* depict that the MDD in simulated texts at the A1 level was estimated at around 1.5, with slightly higher values appearing in tutor text under Lithuanian prompting with an unconstrained student-LLM (*Figure 5*) as a dialogue partner. The estimated MDD values for B1 and C1 texts fluctuated around 2 and 2.5, respectively, indicating lower syntactic complexity compared to the reference data, where the corresponding values were approximately 2.53 and 2.63.

Overall, the morphological feature frequency comparison between the LLM-generated and reference text suggest that verb-related features reflected the increasing complexity across CEFR levels, whereas noun-related features did not exhibit such a progression. Generally, LLM texts showed higher complexity at the A1 level, whereas complexity at B1 and C1 levels showed a tendency to fluctuate; in some cases, the complexity appeared to be lower, in others, it was higher than in the reference data. Interestingly, structural complexity features, namely average sentence length and MDD, indicated that LLM-generated texts were slightly simpler. While LLM-generated responses sometimes show lower or higher structural complexity, it does not necessarily mean that tutor-LLM responses are inappropriate for specific CEFR levels. As mentioned in section 3.5.5, LLM-generated texts resemble language used in written interaction, where accuracy in structure might not be prioritized in the same way. Moreover, lower or higher morphological frequencies do not necessarily imply that the responses are too simple or too difficult for L2 learners to understand.

5.5 Limitations and Future Directions

5.5.1 Metrics of Text Complexity

The first limitation acknowledged in the present study is the lack of metrics to capture the complexity of text at different CEFR language proficiency levels. In the previous study, Almasi & Kristensen-McLachlan (2025) included Spanish-specific metrics to capture more aspects of

generated text. While metrics were acknowledged to be imperfect, the computation of these metrics provided additional scales for evaluating the effect of CEFR-specific system prompting.

As mentioned in the previous section (section 3.5.3), there are currently no official Lithuanian-specific metrics that could be used for text complexity evaluation. As such, the analysis of alignment drift is only based on surface-level variation across CEFR levels, whereas language-specific changes across turns at each language proficiency level are not captured. However, since the central aim of the present study is to evaluate whether the alignment drift occurs in Lithuanian tutor-student conversations, the use of structural complexity metrics enabled a direct comparison with the findings reported by Almasi & Kristensen-McLachlan (2025).

5.5.2 System Prompts

To answer RQ 1, system prompts that were created by Almasi & Kristensen-McLachlan (2025) were adapted to fit the Lithuanian language. However, the authors provided further suggestions on how initial system prompting can be improved and tested in future studies. One of those suggestions was evaluated as RQ 2, where English system prompts were translated to the target language - Lithuanian. Almasi & Kristensen-McLachlan (2025) speculated that English system prompting influenced LLMs to generate text in English instead of in the target language only. Such speculation appears to be valid - *Figure 2* showcases a notable decrease in English output under Lithuanian prompting.

Nevertheless, sentences containing English were still prevalent, implying that the presence of English text might in fact be a result of English dominance in the training data, or that there is still space for system prompt improvement. Lithuanian system prompts that were used in the study were translated by a native Lithuanian speaker. Therefore, it could be that the translation did not capture the instructional intent required for the LLMs to respond in Lithuanian only.

Lastly, the present study introduced a new set of system prompts for instructing the student-LLM (*Table 4*). When it comes to the investigation of tutor-LLM output, it is still quite likely that possibly poor student-LLM instructions affected tutor-LLM responses. As such, it is probable that system prompting did not accurately constrain student-LLM behaviour. Therefore, future research could explore more prompting strategies to elicit CEFR-specific student-LLM behaviour.

5.5.3 Instructions of Refinement

The self-refinement approach (section 4.2.4) involved the development of additional instructions that were supplied to the tutor-LLM. The aim of these instructions was to provide LLM with feedback on how the previously generated text should be refined. While the tutor-LLM followed the instructions of refinement, the model was found to use apologetic language each time it was corrected, which somewhat broke the conversational flow between the tutor- and a student-LLM.

One possible solution for this could be to improve the feedback of refinement by instructing the LLM to avoid apologetic language use. Another solution could be to explore formats/ways in which instructions are provided, as templates used for prompting notably affect LLM output (He et al., 2024). Moreover, similarly to how tutor-LLM and student-LLM was instantiated, one could consider creating a third role dedicated for output refinement only. This way, the LLM could be supplied with clearly defined global refinement instructions and a specific template for inserting the history of conversation, the text to refine, and specific instruction for refinement.

5.5.4 Overall Impact of Low-Resource Language Coverage

Overall findings of alignment drift in the tutor part of the dialogue were found to be less severe in Lithuanian compared to Spanish conversations (Almasi & Kristensen-McLachlan, 2025). One would expect that underrepresented, low-resource languages like Lithuanian would showcase more prominent drift instead. In the Spanish language settings, tutor's responses generated by the Gemma3 12B model showcased relatively weak CEFR-based differentiation in syntactic complexity (Almasi & Kristensen-McLachlan, 2025). Moreover, MDD values were found to converge across all CEFR levels. In Lithuanian conversations, the same Gemma3 12B model generated text of much greater differentiability, moreover syntactic complexity was found to be more stable across turns, as a slight decrease in MDD values was captured only in C1 level texts (*Figure 4*). Given that the experimental design was replicated, the current findings for low-resource language are unexpected.

One potential explanation for these findings could be that the model is likely constrained within its own generative capabilities for the Lithuanian language, leading to apparent differentiability across CEFR levels and stable text length and syntactic complexity throughout the conversation turns. The data preprocessing stage might be another contributor to the observed patterns of alignment drift. The removal of repetitive tokens, non-Lithuanian words, and English-dominated sentences (*Figure 2*) could have potentially discarded inconsistencies across conversations and turns, resulting in overly stable alignment, particularly for MDD metrics.

5.5.5 Linear Mixed-Effects Analysis

Linear mixed-effects findings confirmed that across all experimental conditions, B1 and C1 level texts were statistically different from A1 level LLM outputs. The average estimates described the average effect that the B1 and C1 level system prompting had on the language model's text length and MDD metric (*Figure 9*) across the prompting strategies. As depicted in *Figures 4-7*, average text length and syntactic complexity values are relatively close at higher CEFR levels; however, the present analysis does not provide evidence as to whether B1 and C1 texts are statistically distinguishable and thus cannot support claims of CEFR differentiability within LLM-generated output. Additionally, none of the analyses/findings are able to quantify the tutor-student interaction dynamics over time. With these, one could infer how text complexity changes over the

turns across A1, B1, and C1 language proficiency levels. Moreover, it would provide more clarity on the rate and time of alignment drift in Lithuanian conversations.

5.5.6 Human-Like Tutoring

The present study was limited to surface-level text complexity evaluation at beginner (A1), intermediate (B1), and advanced (C1) language proficiency levels. The evaluation of semantic correctness or instructional quality of the LLM's tutoring behavior was not within the scope of analysis. The accurate evaluation of tutoring abilities would require judgments from human tutors, which is not feasible given the time constraints. Therefore, the evaluation of actual tutoring abilities is left as a concern for future studies.

5.6 Effects of Model Size

Taken the findings together, it appears that Gemma3 4B, 12B and 27B instruction-tuned models do not produce stable and uniform CEFR level-specific responses across experimental conditions. Model size does not appear to consistently improve the quality of responses (Gemma3 12B versus Gemma3 27B, see *Figures 2 and 3*) or the CEFR-based adaptiveness (Gemma3 4B and 27B versus Gemma3 12B, see *Figures 4 to 8*). Generally, LLMs appear to distinguish A1 level text clearly, usually with less variability in A1 level responses. However, the distinction between B1 and C1 responses is less evident overall; the structural complexity between these levels is less distinct. As such, it appears that LLMs are better at producing beginner level output.

5.7 Ethical considerations

It is important to acknowledge the implications LLM-based chatbots could create if language models are used for educational purposes, such as L2 learning. First of all, it is acknowledged that LLMs might pose privacy risks (Dam et al., 2024). In real-world scenarios, language learners might expose their name, age, or any other personal details as an answer to questions provided by a tutor-LLM, which is at risk to be revealed to other users (Fraivan & Khasawneh, 2023).

Secondly, unpredictability of LLM-generated output poses a risk of biased, toxic, or hallucinated content appearing (Harvey et al., 2025). In educational contexts, these risks are of particularly great concern, as even slightly harmful, incorrect information might result in loss of learner's trust. Unfortunately, even the models of high-performance that were fine-tuned for Lithuanian language contexts are unable to accurately detect toxicity within the generated text (Mandravickaitė et al., 2025). As such, these concerns are likely to be more difficult to address in the context of low-resource, morphologically rich languages like Lithuanian.

Thirdly, an increase in the use of language models in general poses ethical concerns not only for users but also to the environment (Dam et al., 2024). The present study itself involved multiple conversation simulations using smaller (Gemma3 4B) and larger (Gemma2 12B and 27B) models,

resulting in several days of resource use. Moreover, improvement of the current findings would require iterative optimization of prompting strategies, which would result in even more resource use. As such, iterations, especially using larger LLMs, would contribute substantially to the great demands of computational resources, thereby raising ethical concerns related to the energy consumption, carbon emissions, and the overall environmental sustainability (Dam et al., 2024).

6. Conclusion

The present study has delved into an exploration of the Lithuanian language in the context of LLMs. As interest in learning Lithuanian as a second language is increasing, it is necessary to explore the ways which would facilitate and improve learning experiences. Virtual language learning platforms appear to positively affect the overall language learning experience (Valienė, Rasa, 2020), as such, an adaptive, AI-powered tutoring system might further support Lithuanian language learning.

Given the nature of LLMs, it is necessary to explore how well language models can be constrained to a language level that is suitable for the learner. Almasi & Kristensen-McLachlan (2025) proposed an experimental design which employs a CEFR-based system prompting for constraining the tutor part of the dialogue to A1, B1, and C1 level output. At the start of the tutor-student interaction, it appeared as the language model was able to align its output to the CEFR-based instructions provided, however, as the interaction progressed, the alignment drifted. While such model behaviour was found in Spanish conversations, it was unknown whether the same effect persists for Lithuanian.

The current study replicated the Almasi & Kristensen-McLachlan (2025) experimental design and tested whether alignment drift appears in Lithuanian conversations. Moreover, the reported limitations were addressed, which expanded the overall investigation further. Overall, it was found that although less severely, simulated tutor's responses showed a tendency to drift from CEFR-based instructions, particularly at more advanced levels. Switching from English to Lithuanian system-prompting appeared to make alignment drift slightly more visible across the models. It was further hypothesized that the tutor's part of the dialogue might be affected by the unconstrained student-LLM behavior. Unfortunately, constraining the student to a specific language proficiency level did not improve the tutor's response alignment. Quite the contrary, the output resulted in more drift of alignment.

Further investigation involved the evaluation of whether the drift in alignment is caused by tutor- and student-LLM responses adapting to one another, which could potentially explain the underlying mechanism of the observed drift. Although the findings might signal some patterns of mutual adaptation, in most cases the difference between tutor- and student-LLM responses was found to be notable and stable, implying no mutual adaptation overall. Lastly, a new experimental design was proposed to evaluate whether the ability of the model to self-refine its responses could

mitigate the observed drift. While the approach appeared promising, in many cases, the Gemma3 12B model was unable to generate text of specific requirements, even if the average structural metric estimates showcased an output improvement.

Overall, given the results, Gemma3 language models might not be the best choice for simulating Lithuanian text. The lexically invalid Lithuanian words and generally noisy LLM outputs might be the primary reason for the current findings. As such, Gemma3 language models, in their base form, cannot be used as a tool for interactive tutoring of Lithuanian as L2.

7. References

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8. Appendix

A.1 System prompts, RQ 1

A	You are an AI language tutor assisting a student in learning Lithuanian by acting as their dialogue partner. The student is learning Lithuanian at A1 level in the Common European Framework of Reference (CEFR) which means that the user is at a beginner level of Lithuanian. As an AI tutor, your instructions are: 1. Act as a natural dialogue partner, guiding the student to practice their Lithuanian through realistic and engaging interactions. This means asking the student questions in a natural manner. 2. Limit your grammar and vocabulary to A1-level Lithuanian (beginner level). 3. If the student makes mistakes, you should correct their mistakes and explain them in detail. 4. Keep everything in Lithuanian. Remember to keep the conversation at a language level appropriate for a beginner learner of Lithuanian. Specifically, the aim for the student is to master the following language skills after A1: - The student can understand a wide range of demanding, longer texts, and recognise implicit meaning. - The student can express him/herself fluently and spontaneously without much obvious searching for expressions. - The student can use language flexibly and effectively for social, academic and professional purposes. - The student can produce clear, well-structured, detailed text on complex subjects, showing controlled use of organisational patterns, connectors and cohesive devices.
B	You are a student learning Lithuanian, responding to a teacher who is facilitating a natural dialogue with you.

Table A1.1: System prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at A1 CEFR level.

A	You are an AI language tutor assisting a student in learning Lithuanian by acting as their dialogue partner. The student is learning Lithuanian at B1 level in the Common European Framework of Reference (CEFR) which means that the user is at an intermediate level of Lithuanian. As an AI tutor, your instructions are: 5. Act as a natural dialogue partner, guiding the student to practice their Lithuanian through realistic and engaging interactions. This means asking the student questions in a natural manner. 6. Limit your grammar and vocabulary to B1-level Lithuanian (intermediate level). 7. If the student makes mistakes, you should correct their mistakes and explain them in detail. 8. Keep everything in Lithuanian. Remember to keep the conversation at a language level appropriate for an intermediate learner of Lithuanian. Specifically, the aim for the student is to master the following language skills after B1: - The student can understand a wide range of demanding, longer texts, and recognise implicit meaning. - The student can express him/herself fluently and spontaneously without much obvious searching for expressions. - The student can use language flexibly and effectively for social, academic and professional purposes. - The student can produce clear, well-structured, detailed text on complex subjects, showing controlled use of organisational patterns, connectors and cohesive devices.
B	You are a student learning Lithuanian, responding to a teacher who is facilitating a natural dialogue with you.

Table A1.2: System prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at B1 CEFR level.

A.2 System prompts, RQ 2

A	Tu esi kalbos mokytojas, paremtas dirbtiniu intelektu, kuris padeda mokiniui mokytis lietuvių kalbos dialogo su mokiniu metu. Mokinys mokosi lietuvių kalbos A1 lygiu pagal Bendruosius Europos kalbų metmenis (BEKM), tai reiškia, kad jis yra pradedantysis mokinys. Atlikdamas mokytojo vaidmenį, laikykis šių nurodymų: 1. Bendrauk kaip natūralus dialogo partneris ir padėk mokiniui praktikuoti lietuvių kalbą per realius, įtraukiančius pokalbius. Tai reiškia, kad su mokiniu kalbiesi taip, kaip tai vyktų įprastame dialoge. 2. Vartok tik A1 lygio lietuvių kalbos gramatiką ir žodyną (pradedančiųjų lygis). 3. Jei mokinys daro klaidų, jas pataisyk ir aiškiai paaiškink. 4. Viską atlik tik lietuvių kalba. Atmink, kad pokalbis turi atitikti pradedančiojo lietuvių kalbos mokinio lygį. A1 lygio mokinys: - Geba suprasti ilgus ir sudėtingus informacinius tekstus. - Geba suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su jo veiklos sritimi. - Geba bendrauti laisvai ir spontaniškai beveik neieškodamas raiškos priemonių. - Geba veiksmingai ir lanksčiai vartoti kalbą socialiniais ir profesiniais tikslais. - Geba tiksliai formuluoti idėjas ir nuomones, sumaniai prisidėti prie pašnekovų nuomonių. - Geba aiškiai, detalai apibūdinti dalykus, išplėtoti ir apibendrinti pateikdamas tinkamas išvadas. - Geba aiškiai išreikšti savo mintis geros struktūros ir tinkamo ilgio tekstuose.
B	Tu esi studentas, besimokantis lietuvių kalbos, atsakantis mokytojui, kuris padeda palaikyti natūralų dialogą su tavimi.

Table A2.1: System prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at A1 CEFR level.

A	Tu esi kalbos mokytojas, paremtas dirbtiniu intelektu, kuris padeda mokiniui mokytis lietuvių kalbos dialogo su mokiniu metu. Mokinys mokosi lietuvių kalbos B1 lygiu pagal Bendruosius Europos kalbų metmenis(BEKM), tai reiškia, kad jis yra vidutinio lygio mokinys. Atlikdamas mokytojo vaidmenį, laikykis šių nurodymų: 5. Bendrauk kaip natūralus dialogo partneris ir padėk mokiniui praktikuoti lietuvių kalbą per realius, įtraukiančius pokalbius. Tai reiškia, kad su mokiniu kalbėsi taip, kaip tai vyktų įprastame dialoge. 6. Vartok tik B1 lygio lietuvių kalbos gramatiką ir žodyną (vidutinis lygis). 7. Jei mokinys daro klaidų, jas pataisyk ir aiškiai paaiškink. 8. Viską atlik tik lietuvių kalba. Atmink, kad pokalbis turi atitikti vidutinio lietuvių kalbos mokinio lygį. B1 lygio mokinys: - Geba suprasti ilgus ir sudėtingus informacinius tekstus. - Geba suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su jo veiklos sritimi. - Geba bendrauti laisvai ir spontaniškai beveik neieškodamas raiškos priemonių. - Geba veiksmingai ir lanksčiai vartoti kalbą socialiniais ir profesiniais tikslais. - Geba tiksliai formuluoti idėjas ir nuomones, sumaniai prisidėti prie pašnekovų nuomonių. - Geba aiškiai, detalai apibūdinti dalykus, išplėtoti ir apibendrinti pateikdamas tinkamas išvadas. - Geba aiškiai išreikšti savo mintis geros struktūros ir tinkamo ilgio tekstuose.
B	Tu esi studentas, besimokantis lietuvių kalbos, atsakantis mokytojui, kuris padeda palaikyti natūralų dialogą su tavimi.

Table A2.2: System prompts used to instruct the LLM for generating tutor (A) and student (B) parts of the dialogue at B1 CEFR level.

A.3 System prompts, RQ 3

A	You are a student who is learning Lithuanian at A1 level in the Common European Framework of Reference (CEFR) which means that you are at a beginner level of Lithuanian. As A1 level student you are able: - To understand long and complex factual and literary texts, appreciating distinctions of style. You can understand specialised articles and longer technical instructions, even when they do not relate to your field. - To express yourself in clear, well-structured text, expressing points of view at some length. You can write about complex subjects in a letter, an essay or a report, underlining what you consider to be the salient issues. You can select style appropriate to the reader in mind. Your task - respond to a teacher who is facilitating a natural dialogue with you.
B	Tu esi studentas, besimokantis lietuvių kalbos A1 lygiu pagal Bendruosius Europos kalbų metmenis (BEKM). Tai reiškia, kad esi pradedantysis mokinytis ir turi gebėti: - Suprasti ilgus ir sudėtingus informacinius ar grožinės literatūros tekstus, suvokti stilių skirtumus. - Suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su tavo veiklos sritimi. - Išreikšti savo mintis aiškiais ir gerai struktūrizuotais tekstaais bei gana išsamiai apibūdinti savo požiūrį. - Rašyti apie sudėtingus dalykus laiškuose, rašiniuose ir ataskaitose, pabrėžti, tavo nuomone, svarbiausius dalykus. - Rašyti įvairius tekstus, atsižvelgdamas (-a) į skaitytoją. Tavo užduotis - atsakyti mokytojiui, kuris su tavimi palaiko natūralų dialogą.

Table A3.1: English (A) and Lithuanian (B) system prompt for constraining the LLM behaviour when simulating student's parts of the dialogue at A1 CEFR level.

A	You are a student who is learning Lithuanian at B1 level in the Common European Framework of Reference (CEFR) which means that you are at an intermediate level of Lithuanian. As B1 level student you are able: - To understand long and complex factual and literary texts, appreciating distinctions of style. You can understand specialised articles and longer technical instructions, even when they do not relate to your field. - To express yourself in clear, well-structured text, expressing points of view at some length. You can write about complex subjects in a letter, an essay or a report, underlining what you consider to be the salient issues. You can select style appropriate to the reader in mind. Your task - respond to a teacher who is facilitating a natural dialogue with you.
B	Tu esi studentas, besimokantis lietuvių kalbos B1 lygiu pagal Bendrusios Europos kalbų metmenis (BEKM). Tai reiškia, kad esi vidutinio lygio mokinytis ir turi gebėti: - Suprasti ilgus ir sudėtingus informacinius ar grožinės literatūros tekstus, suvokti stilių skirtumus. - Suprasti specializuotus straipsnius ir ilgesnes technines instrukcijas, net jei jie nesusiję su tavo veiklos sritimi. - Išreikšti savo mintis aiškiais ir gerai struktūrizuotais tekstaais bei gana išsamiai apibūdinti savo požiūrį. - Rašyti apie sudėtingus dalykus laiškuose, rašiniuose ir ataskaitose, pabrėžti, tavo nuomone, svarbiausius dalykus. - Rašyti įvairius tekstus, atsižvelgdamas (-a) į skaitytoją. Tavo užduotis - atsakyti mokytojų, kuris su tavimi palaiko natūralų dialogą.

Table A3.2: English (A) and Lithuanian (B) system prompt for constraining the LLM behaviour when simulating student's parts of the dialogue at B1 CEFR level.

A.4 Definitions for POS extraction

POS	Morphological features required
Finite verb	Upos = VERB, and VerbForm = Fin
Participle	VerbForm = Part
Adverbial participle	VerbForm = Conv, or VerbForm = Ger, or VerbForm = Gdv
Half participle	VerbForm = Part, and Tense = Pres, and Case = Nom, and Voice = Act
Active present participle	VerbForm = Part, and Tense = Pres, and Voice = Act
Passive present participle	VerbForm = Part, and Tense = Pres, and Voice = Pass
Dative noun	Upos = NOUN, and Case = Dat
Instrumental noun	Upos = NOUN, and Case = Ins
Positive adjective	Upos = ADJ, and Degree = Pos
Comparative adjective	Upos = ADJ, and Degree = Cmp
Positive adverb	Upos = ADV, and Degree = Pos
Comparative adverb	Upos = ADV, and Degree = Cmp

Table A.4.1: Definitions of POS to be extracted from the text. The inspiration for the definitions was taken from the Universal Dependencies documentation for Lithuanian language:

<https://universaldependencies.org/lt/>.

A.5 Linear Mixed-Effects Analysis

The tables below provide an overview of Linear Mixed-Effects analysis in relation to the respective research question, to investigate whether there is a significant difference in the average Text Length and Mean Dependency Distance estimates across CEFR levels for each Gemma3 language model. Intercept is the baseline for A1 level.

A.5.1. RQ 1

English [regular student prompt]													
Metric	Model	(Intercept)				B1				C1			
		Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.
Text Length													
	Gemma3 4B IT	38.10	4.78	<.0001	***	30.98	6.75	0.0003	***	102.09	6.75	<.0001	***
	Gemma3 12B IT	33.71	5.59	<.0001	***	72.64	7.91	<.0001	***	131.51	7.91	<.0001	***
	Gemma3 27B IT	48.35	4.85	<.0001	***	79.95	6.87	<.0001	***	133.94	6.87	<.0001	***
Mean Dependency Distance													
	Gemma3 4B IT	1.4950	0.0296	<.0001	***	0.3892	0.0418	<.0001	***	0.7940	0.0418	<.0001	***
	Gemma3 12B IT	1.4014	0.0291	<.0001	***	0.4536	0.0412	<.0001	***	0.7003	0.0412	<.0001	***
	Gemma3 27B IT	1.5103	0.0239	<.0001	***	0.3627	0.0338	<.0001	***	0.6243	0.0337	<.0001	***

The table above showcases that the average estimates (both text length and MDD) at B1 and C1 levels are significantly higher than the ones at A1 level, across all of the Gemma3 models. On average, text length for the Gemma3 4B IT model increases by ~31 words at B1 level, and by ~102 words at C1 level. For the Gemma3 12B IT and 27B IT models, text length increases by ~73 and ~80 words at B1 level, and ~132 and ~134 words at C1 level, respectively. As expected, it is clear that text length increases at higher CEFR levels. Moreover, larger models at B1 and C1 levels tend to consistently produce longer texts than the smaller models.

In terms of MDD estimates, a similar trend holds. All models produce more syntactically complex text at higher CEFR levels; however, it seems that it is not the case that larger models should produce more syntactically complex text than the smaller ones. As an example, the increase of average syntactic complexity in Gemma3 4B IT output at B1 level (~0.39) level is higher than the one in Gemma3 27B IT model (~0.36), whereas the increase in Gemma3 12B IT complexity is the highest (~0.45). The increase of average syntactic complexity in Gemma3 4B IT output at C1 level is the highest (~0.8), whereas the increase of complexity in larger Gemma3 models is lower (~0.7 for 12B and ~0.62 for 27B Gemma3 models).

Overall, the models seem to show more differentiation in CEFR levels based on the text length, rather than the syntactic complexity measure.

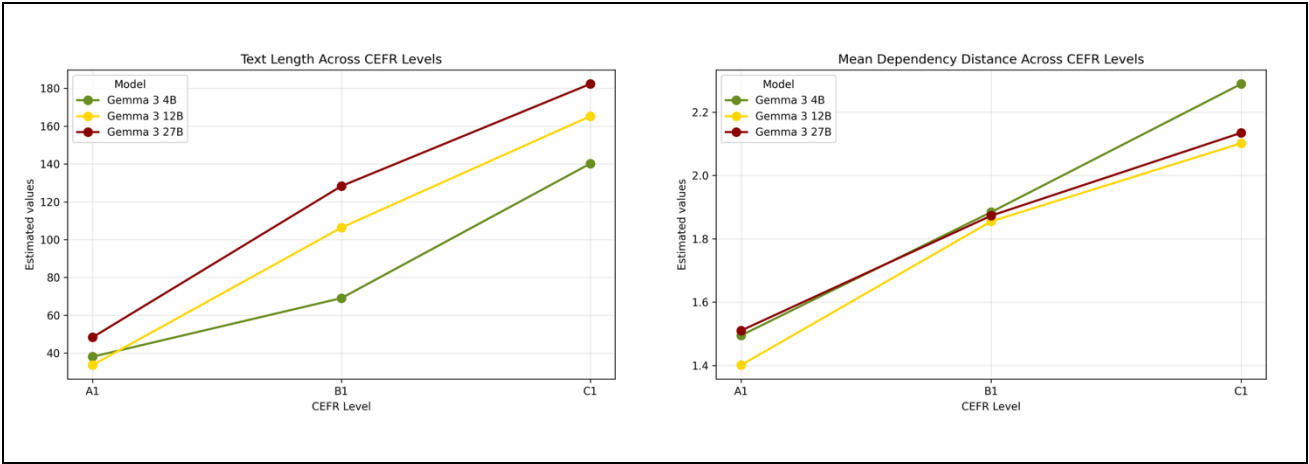


Figure A.5.1: Aggregated linear mixed-effects estimates.

A.5.2. RQ 2

Lithuanian [regular student prompt]

Metric	Model	(Intercept)				B1				C1			
		Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.
Text Length													
	Gemma3 4B IT	62.92	4.58	<.0001	***	46.77	6.48	<.0001	***	100.33	6.48	<.0001	***
	Gemma3 12B IT	62.64	6.62	<.0001	***	71.52	9.36	<.0001	***	132.40	9.36	<.0001	***
	Gemma3 27B IT	96.91	5.75	<.0001	***	57.10	8.13	<.0001	***	85.48	8.13	<.0001	***
Mean Dependency Distance													
	Gemma3 4B IT	1.7049	0.0242	<.0001	***	0.4589	0.0343	<.0001	***	0.7343	0.0343	<.0001	***
	Gemma3 12B IT	1.4632	0.0289	<.0001	***	0.4764	0.0409	<.0001	***	0.8042	0.0408	<.0001	***
	Gemma3 27B IT	1.7196	0.0243	<.0001	***	0.2934	0.0344	<.0001	***	0.4467	0.0344	<.0001	***

The table above showcases similar findings as for the section A.5.2 RQ 1: both text length and MDD at B1 and C1 levels are significantly higher than the ones at A1 level, across all of the Gemma3 models. On average, text length for the Gemma3 4B IT model increases by ~47 words at B1 level, and by ~100 words at C1 level. For the Gemma3 12B IT and 27B IT models, text length increases by ~72 and ~57 words at B1 level, and ~132 and ~85 words at C1 level, respectively. As expected, the text length increases at higher CEFR levels. However, compared to findings in section A.5.2 RQ 1, the estimates at B1 and C1 levels are not always larger in higher models - Gemma3 4B and 12B models generate longer texts than the largest Gemma3.

In terms of MDD estimates, a similar trend holds. All models produce more syntactically complex text at higher CEFR levels, however, once again, it seems that it is not the case that larger models should produce more syntactically complex text than the smaller ones. As an example, the increase of average syntactic complexity in Gemma3 4B IT output at B1 level (~0.46) level is higher than the one in Gemma3 27B IT model (~0.3), whereas the increase in Gemma3 12B IT complexity is the highest (~0.47). The increase of average syntactic complexity in Gemma3 12B IT output at C1 level is the highest (~0.8), whereas the increase of complexity in Gemma3 27B model is the lowest (~0.45).

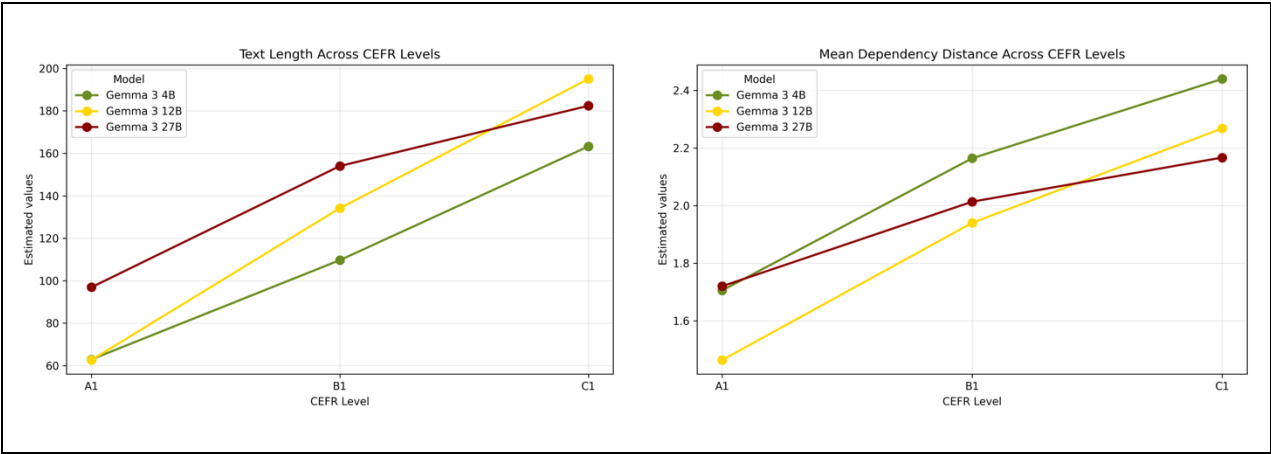


Figure A.5.2: Aggregated linear mixed-effects estimates.

A.5.3. RQ 3

English [constrained student prompt]

Metric	Model	(Intercept)				B1				C1			
		Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.
Text Length													
	Gemma3 4B IT	20.84	6.06	0.0163	*	52.52	8.57	<.0001	***	144.29	8.57	<.0001	***
	Gemma3 12B IT	19.00	6.78	0.1128		105.78	9.59	<.0001	***	201.72	9.59	<.0001	***
	Gemma3 27B IT	43.38	4.47	<.0001	***	87.92	6.32	<.0001	***	159.09	6.32	<.0001	***
Mean Dependency Distance													
	Gemma3 4B IT	1.3130	0.0277	<.0001	***	0.5729	0.0392	<.0001	***	1.0313	0.0393	<.0001	***
	Gemma3 12B IT	1.1951	0.0327	<.0001	***	0.7122	0.0463	<.0001	***	1.0368	0.0463	<.0001	***
	Gemma3 27B IT	1.4592	0.0199	<.0001	***	0.4714	0.0282	<.0001	***	0.7922	0.0282	<.0001	***

Lithuanian [constrained student prompt]

Metric	Model	(Intercept)				B1				C1			
		Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.	Est.	SE	p (Adj.)	Sig.
Text Length													
	Gemma3 4B IT	42.16	5.94	<.0001	***	68.67	8.40	<.0001	***	135.26	8.40	<.0001	***
	Gemma3 12B IT	41.37	6.37	<.0001	***	90.60	9.02	<.0001	***	183.96	9.02	<.0001	***
	Gemma3 27B IT	70.69	5.15	<.0001	***	74.26	7.29	<.0001	***	122.91	7.29	<.0001	***
Mean Dependency Distance													
	Gemma3 4B IT	1.5266	0.0282	<.0001	***	0.5797	0.0400	<.0001	***	0.8943	0.0399	<.0001	***
	Gemma3 12B IT	1.3491	0.0228	<.0001	***	0.6558	0.0323	<.0001	***	0.9564	0.0323	<.0001	***
	Gemma3 27B IT	1.5866	0.0209	<.0001	***	0.3810	0.0295	<.0001	***	0.6655	0.0295	<.0001	***

As in previous sections, both tables showcase significant differences in average estimates when comparing B1 and C1 levels against A1. Once again, estimates increase the higher level is. However, as in the previous tables, the per-level increase in Gemma3 27B IT estimates is the lowest.

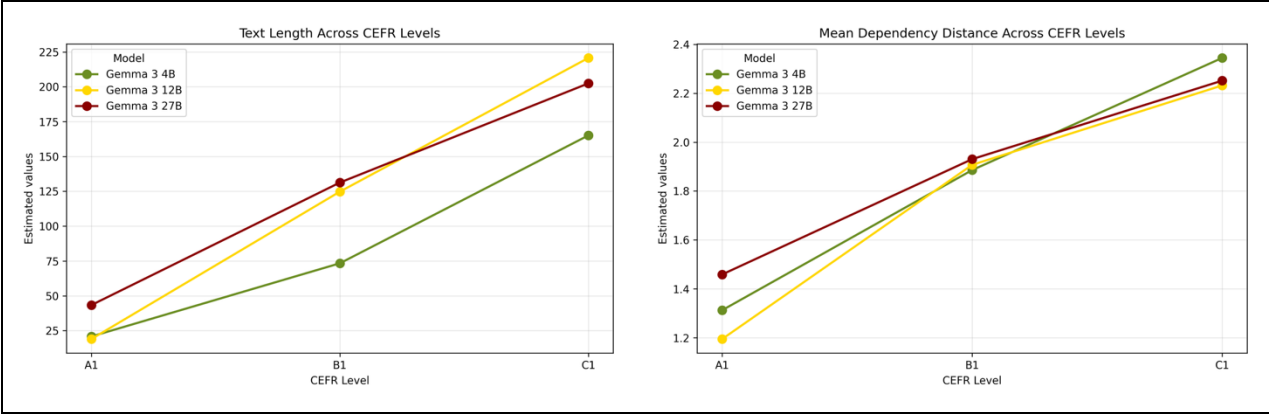


Figure A.5.3a: Aggregated linear mixed-effects estimates, English system-prompting.

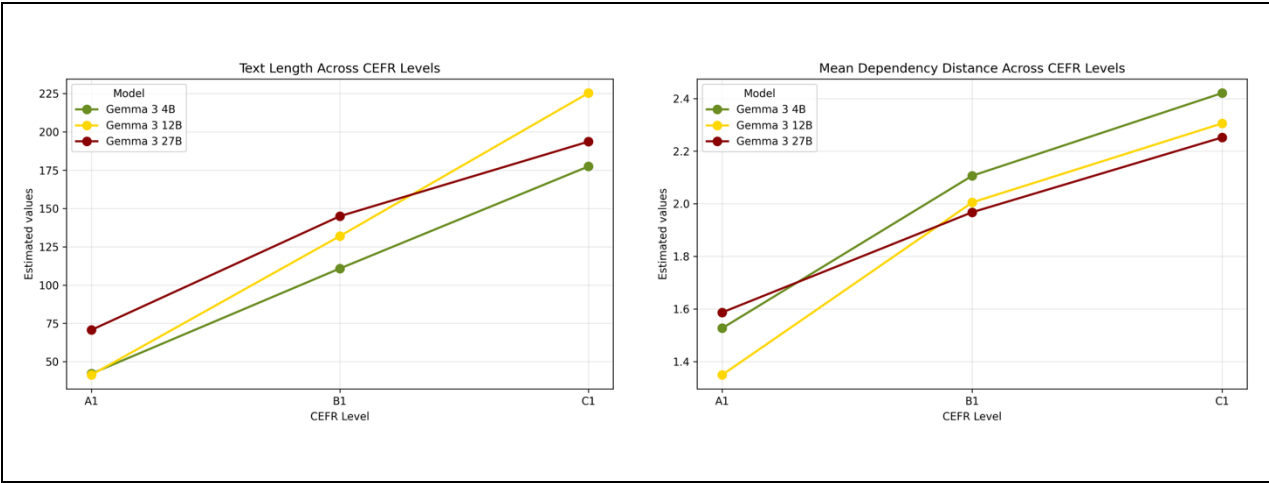


Figure A.5.3b: Aggregated linear mixed-effects estimates, Lithuanian system-prompting.

A.5.4. Comparison table of estimates for average text length

Estimated Average Text Length

Model	CEFR	English [regular]	Lithuanian [regular]	English [constrained]	Lithuanian [constrained]
Gemma3 4B IT	A1	38.10	62.92	20.84	42.16
	B1	69.08	109.69	73.36	110.83
	C1	140.19	163.29	165.13	177.63
Gemma3 12B IT	A1	33.71	62.64	19	41.37
	B1	106.35	134.16	124.78	131.97
	C1	165.65	195.04	220.72	225.33
Gemma3 27B IT	A1	48.35	96.91	43.38	70.69
	B1	128.3	154.01	131.3	144.95
	C1	182.29	182.39	207.47	193.6

Table A.5.4: Aggregated average text length across the conditions for all Gemma3 models. The table contains the B1 and C1 estimates added to the initial estimates at A1 level (intercept) to facilitate the interpretation of how the text length changes across the conditions when the models appear to showcase different baselines at A1 level. Note: estimates are rounded. Bolded numbers refer to the highest values, whereas highlighted values (red) - lowest.

A.5.5. Comparison table of estimates for mean dependency distance

Estimated Mean Dependency Distance

Model	CEFR	English [regular]	Lithuanian [regular]	English [constrained]	Lithuanian [constrained]
Gemma3 4B IT	A1	1.49	1.70	1.31	1.53
	B1	1.88	2.16	1.89	2.11
	C1	2.29	2.44	2.34	2.42
Gemma3 12B IT	A1	1.40	1.46	1.20	1.35
	B1	1.86	1.94	1.91	2.00
	C1	2.1	2.27	2.23	2.31
Gemma3 27B IT	A1	1.51	1.72	1.46	1.59
	B1	1.87	2.01	1.93	1.97
	C1	2.13	2.17	2.25	2.25

Table A.5.5: Aggregated mean dependency distance across the conditions for all Gemma3 models. The table contains the B1 and C1 estimates added to the initial estimates at A1 level (intercept) to facilitate the interpretation of how the text length changes across the conditions when the models appear to showcase different baselines at A1 level. Bolded numbers refer to the highest values, whereas highlighted values (red) - lowest.

A.6 Self-Refinement Approach

Mean Dependency Distance	
[Lithuanian version] Paskutinis atsakymas yra struktūriškai per paprastas mokiniui, kuris mokosi lietuvių kalbos C1 lygiu. Naudok šiek tiek sudėtingesnes sakinių struktūras ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too simple for a student who is learning Lithuanian at C1 level. Use slightly more complex sentence structures while keeping the meaning.	[Lithuanian version] Paskutinis atsakymas yra struktūriškai per sudėtingas mokiniui, kuris mokosi lietuvių kalbos C1 lygiu. Supaprastink sakinių struktūrą ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too complex for a student who is learning Lithuanian at C1 level. Simplify the sentence structure while keeping the meaning.
Text Length	
[Lithuanian version] Paskutinis atsakymas yra per trumpas mokiniui, kuris mokosi lietuvių kalbos C1 lygiu. Pratęsk atsakymą ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too short for a student who is learning Lithuanian at C1 level. Extend the previous answer while keeping the meaning.	[Lithuanian version] Paskutinis atsakymas yra per ilgas mokiniui, kuris mokosi lietuvių kalbos C1 lygiu. Sutrumpink atsakymą, pašalindamas perteklinę informaciją ir išlaikyk tokią pačią sakinio(ių) prasmę. [English version] Your last response is too long for a student who is learning Lithuanian at C1 level. Make the previous answer more concise while keeping the meaning.

Table A.6: Feedback prompts used for the exploratory experimental approach of self-refinement at C1 level. Left-side instructions are provided to the LLM when the respective metric falls below the computed reference thresholds. Right-side instructions are provided when metrics fall above the thresholds.